

Chapter 14

Low-Power Management

14.1 Overview

ESP32-P4 features an advanced low-power management system that optimizes the chip's power consumption while maintaining its high performance.

The low-power management system employs various power-saving techniques such as sleep modes, dynamic voltage and frequency scaling, and peripheral power gating to minimize the chip's power consumption.

At the core of this low-power management system is the power management unit (PMU) hardware component. The PMU powers up and down different power domains of the chip to achieve the best balance between chip performance, power consumption, and wake-up latency.

14.2 Terminology

The following terms related to low-power management are defined in the context of the *ESP32-P4 Technical Reference Manual* to help readers better understand this document:

Low-power management	Refers to the whole system that manages the chip's power consumption.
Power management unit (PMU)	Refers to the specific hardware module that controls power up and down for power domains, clocks, and power-related logic.
Power domain	Refers to the smallest unit that can be independently powered up or down. A power domain can contain one or multiple modules within the chip.
PMU states	Refers to the states of the PMU's state machine. Users can configure the clock gating and power gating of a power domain in each of the states.
Power modes	Refers to the preset power modes that power up different domains for typical application scenarios.

14.3 Features

The PMU has the following features:

- Supports three configurable PMU states. Software can flexibly configure them according to the needs.
 - HP_ACTIVE
 - HP_SLEEP

- LP_SLEEP
- Supports five preset power modes that suit various typical usage scenarios:
 - Active
 - Light-sleep0 (This mode does not power down the peripherals, but only the CPU. The clocks are not turned off. The power consumption is high, and the wake-up latency is short.)
 - Light-sleep1 (This mode does not power down the peripherals, but only turns off the clocks. The power consumption is relatively higher and the wake-up latency shorter.)
 - Light-sleep2 (This mode powers down the peripherals and needs data backup and restoration, resulting in lower power consumption and a longer wake-up latency.)
 - Deep-sleep
- LP SRAM (32 KB): Keep power up in all modes, can store HP CPU's non-volatile data, and LP CPU's instructions and data.
- Supports a programmable retention DMA that backs up and restores the CPU and peripherals status when the chip switches between PMU states.
- Supports a power controller that controls the power and clocks depending on the power modes.

14.4 Functional Description

ESP32-P4's low-power management involves the following components:

- **Power scheme:** The power scheme of ESP32-P4 includes power regulators, digital power domains, analog power domains, etc.
- **PMU controller:** It is the core part of the PMU, controlling the power up and down of power domains, clocks, etc.
- **One RTC timer:** For more information, please refer to Chapter [18 RTC Timer](#).
- **16 LP GPIO pins (GPIO0–GPIO15):** These pins are always powered up and are not affected by any low-power modes, which makes them suitable for working as wake-up sources when the chip is in low-power modes. These pins can also work as regular GPIOs. For more information on LP GPIOs, refer to Chapter [9 GPIO Matrix and IO MUX](#).
- **32 KB SRAM:** The 32 KB SRAM is accessible to both HP CPU and LP CPU. It works under the HP CPU clock when accessed by the HP CPU and under the LP CPU clock when accessed by the LP CPU.
- **Brownout detector:** Monitors the power of the voltage supply pins, ensuring stable chip operation and preventing the SoC from potential malfunction if it is subject to glitches or under-voltage. For more information, refer to Chapter [23 Brown-out Detector](#).

The following sections provide a detailed description of the components mentioned above.

14.4.1 Power Scheme

Figure [14.4-1](#) shows the power scheme of ESP32-P4 that mainly includes:

- A DCDC voltage regulation feedback system

- Two internal regulators
- Four output regulators (EXT_LDO)
- Analog power domains
- Digital power domains
- Battery power domain

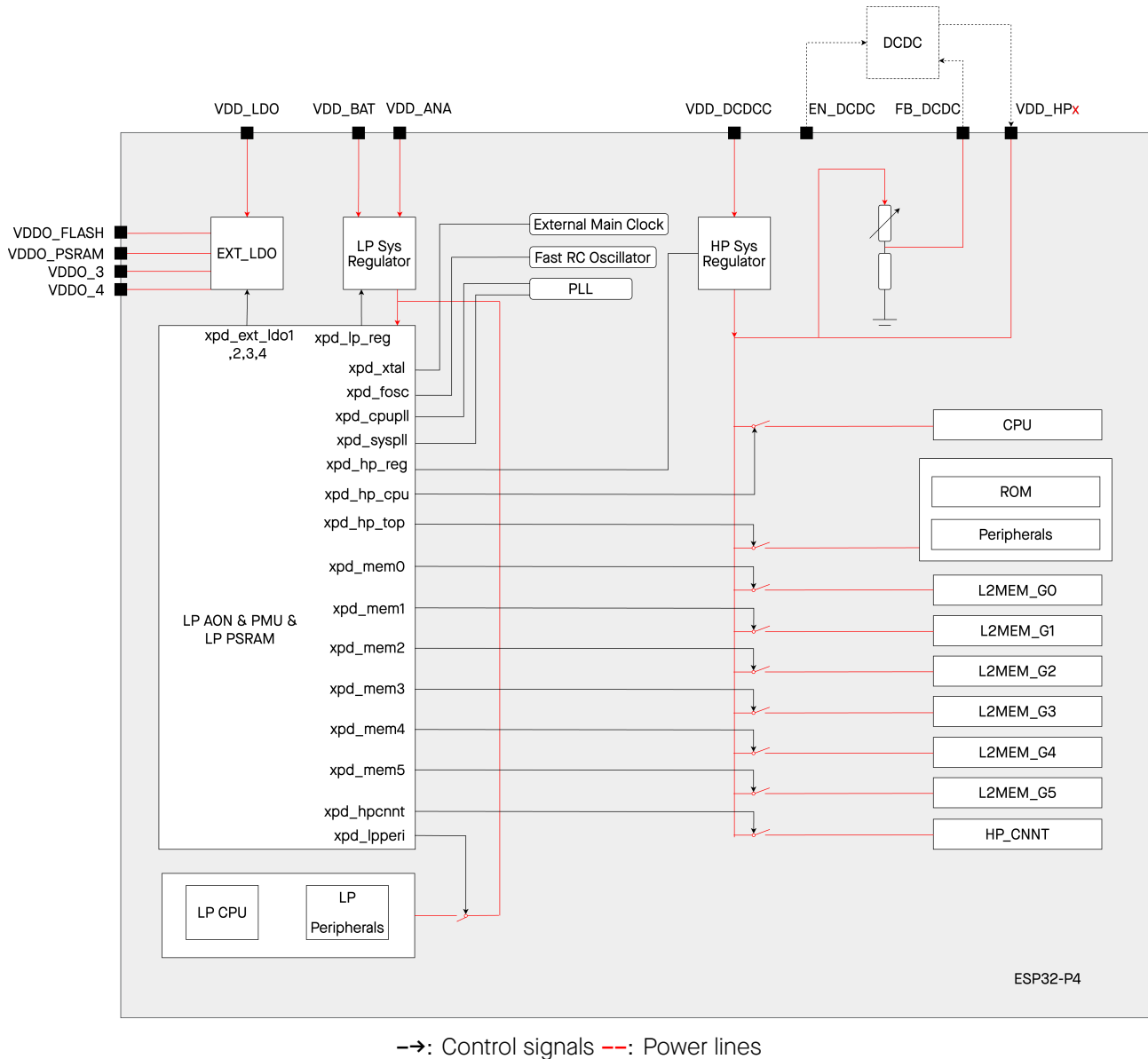


Figure 14.4-1. ESP32-P4 Power Scheme

14.4.1.1 Regulators

As shown in Figure 14.4-1, the analog section of ESP32-P4 includes one external DCDC with feedback regulation and two internal HP/LP system regulators, responsible for regulating the power supply for different power domains.

- The **DCDC voltage regulation feedback system** adjusts the voltage of the external DCDC.

- The **HP (high-performance) system regulator** has strong drive capability, higher power consumption, and adjustable output voltage. It is only suitable for power-up and sleep modes.
- The **LP (low-power) system regulator** regulates the power supply of low-power modules. Its output voltage is also adjustable.

The DCDC voltage regulation feedback system and the HP system regulator are responsible for powering the HP system. The external DCDC can adjust its output voltage through the internal voltage feedback system. By default, the chip is powered by the HP system regulator when powered up. After power-up is complete, it is recommended to switch to the DCDC power supply for better efficiency and load capacity.

14.4.1.2 Digital Power Domains

- The HP system contains the following digital power domains:
 - **CPU (PD_HP_CPU)**: Mainly includes the CPU and peripherals that support its operation.
 - **Peripherals + ROM (PD_TOP)**: Mainly includes the bus and HP peripherals.
 - **L2MEM_G0–G5**: It is divided into six sub-power domains, as shown in Figure 14.4-1.
 - **HP_CNNT (PD_HP_CNNT)**: Mainly includes USB and SDIO modules.

There is an independent power switch between the regulator and each power domain, enabling up/down control of the digital power domain.

- The LP system contains the following digital power domains:
 - **LP PD peripherals (PD_LP_PERI)**: Mainly includes LP CPU and LP peripherals.
 - **LP always-on (PD_AON)**: It mainly includes LP always-on peripherals (e.g., RTC timer) and PMU controller. This power domain keeps powered up all the time.

14.4.1.3 Analog Power Domains

- The HP system contains the following analog power domains:
 - CPLL_CLK
 - SPLL_CLK
 - Audio PLL
 - SDIO PLL
- The LP system contains the following analog power domains:
 - External Main Clock
 - Fast RC Oscillator

14.4.1.4 Battery Power Domain

The battery power domain (VDD_BAT) serves as a backup power domain. When all external power supplies (VDD_ANA) are turned off, the low-power system regulators, LP_SLOW_CLK, and LP_FAST_CLK can all be powered by the VDD_BAT supply.

14.4.2 PMU

The PMU of ESP32-P4 controls power consumption-related components of each power domain, such as power and clock. PMU consists of the following major parts:

- **PMU main state machine:** It records and switches the PMU states.
- **Sleep/wake-up controller:** It sends sleep or wake-up requests to the PMU main state machine.
- **Power controllers:** They control the power and clock signals depending on the power modes of the chip. The power controllers include:
 - **Digital power controller:** It powers up or down the digital power domains.
 - **Analog power controller:** It enables the analog modules, such as the regulators, analog clocks, etc.
 - **Clock controller:** It manages the clock gating of peripherals and selects analog clock sources for digital clocks.
 - **Data backup controller:** It controls the data backup and restore process when the chip switches between PMU states.
 - **System controller:** It controls some system-level modules, such as suspending watchdog functionality in sleep mode (when the CPU is unavailable).

The PMU workflow involves the following steps:

- The sleep/wake-up controller sends sleep or wake-up requests to the PMU main state machine.
- The PMU main state machine generates power gating, clock gating, and reset signals.
- The power controllers and clock controllers power up or down different power domains and clocks based on the signals generated by the PMU main state machine, allowing the chip to enter or exit different low-power modes.

The PMU workflow is shown in Figure [14.4-2](#). The definitions of HP_SWITCH and LP_SWITCH are as follows:

- **HP_SWITCH:** The intermediate state for the HP system transitioning between HP_ACTIVE and HP_SLEEP. The hardware completes the control switching of various controllers in this state.
- **LP_SWITCH:** The intermediate state for the LP system transitioning between HP_SLEEP and LP_SLEEP. The hardware completes the control switching of various controllers in this state.

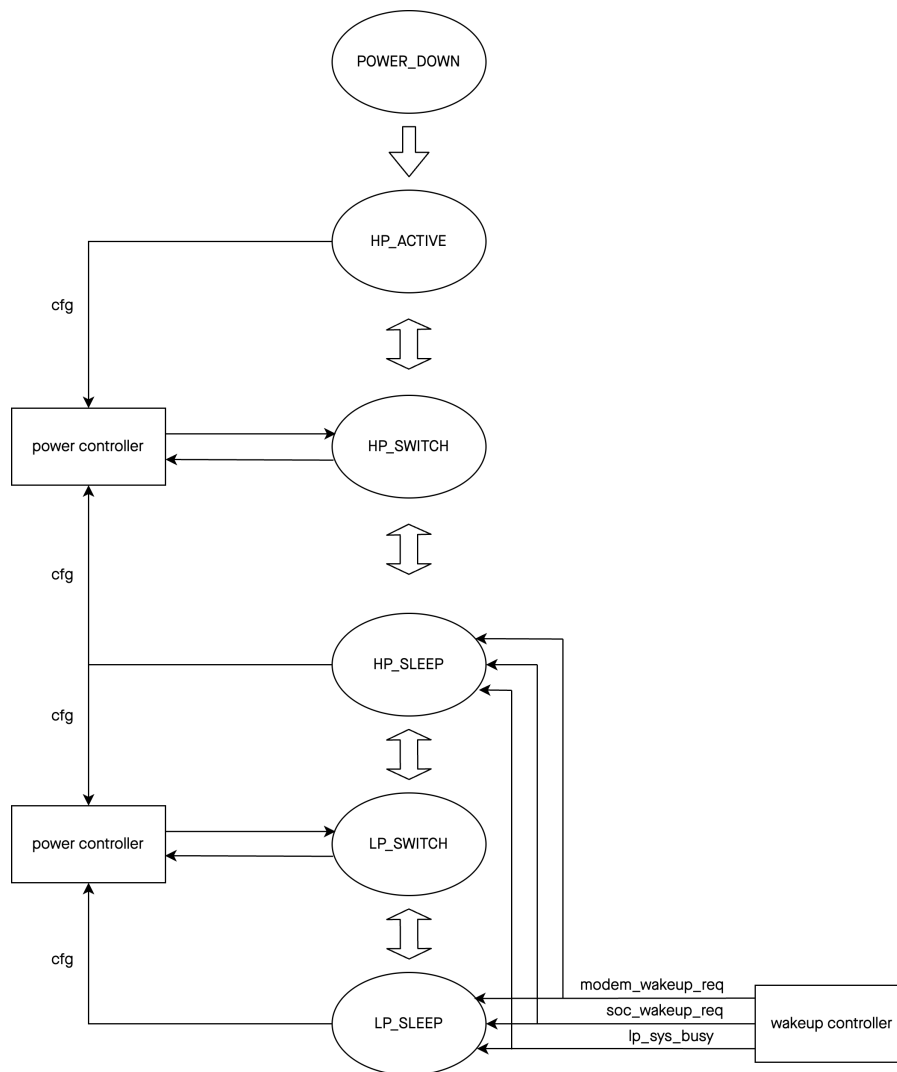


Figure 14.4-2. PMU Workflow

The following sections describe the main parts of PMU.

14.4.2.1 PMU Main State Machine

The PMU main state machine receives sleep and wake-up signals, changes the state of power and clock through the power controllers, thereby switching PMU states, and achieving a balance between performance and power consumption of the chip.

The PMU main state machine supports three PMU states, each controlled by different sleep and wake-up signals, supporting software customization of power and clocks. These PMU states allow the software to expand power modes for various application scenarios. The three PMU states are:

- **HP_ACTIVE:** The circuits on the chip are powered up to a maximum, supporting the HP system and LP system operation.
- **HP_SLEEP:** The HP system is in sleep, supporting LP peripherals operation.
- **LP_SLEEP:** The HP system and LP peripherals are in sleep, while the always-on circuits remain operational.

Note:

The division of the HP and LP systems is as follows:

- LP system (peripherals): LP RISC-V 32-bit Microprocessor, LP GPIO, LP UART, LP I2C, LP I2S, touch sensor, temperature sensor, LP ADC, LP SPI
- LP system (always-on circuits): PMU, RTC Watchdog Timer, Super Watchdog
- HP system: All peripherals except those belonging to the LP system.

For more details, please refer to [ESP32-P4 Datasheet](#) > *Functional Block Diagram*.

HP_ACTIVE is the operating state of the HP system, HP_SLEEP is the sleep state of the HP system, LP_SLEEP is the sleep state of the LP system. The operating state of the LP system reuses the HP_SLEEP state and is therefore also referred to as HP_SLEEP.

- If a module belongs to the HP system, its power up/down can be configured in the HP_ACTIVE/HP_SLEEP states. It reuses the HP_SLEEP state in the LP_SLEEP state.
- If a module belongs to the LP system, its power up/down can be configured in the HP_SLEEP/LP_SLEEP state. It reuses the HP_SLEEP state in the HP_ACTIVE state.

Take the HP CPU as an example. The HP CPU belongs to the HP system, so it can be configured to power up/down in the HP_ACTIVE/HP_SLEEP states through the following registers and reuse the HP_SLEEP configurations in LP_SLEEP.

- HP_ACTIVE: [PMU_HP_ACTIVE_PD_HP_CPU_PD_EN](#)
- HP_SLEEP: [PMU_HP_SLEEP_PD_HP_CPU_PD_EN](#)

Similarly, users can configure the power up and down for other power domains in different PMU states. For the configuration registers, please refer to Section 14.8.

Note:

In the following text, all such configuration registers are collectively referred to as PMU_*n1*_PD_*POWERDOMAIN*_PD_EN, where *n1* represents the three PMU states.

Once the configuration is done, the PMU uses various controllers to make these configurations effective, as described in the following sections.

14.4.2.2 Sleep/Wake-up Controller

The sleep/wake-up controller initiates sleep and wake-up requests to the PMU main state machine. ESP32-P4 supports multiple wake-up sources to wake the CPU from different power modes. Table 14.4-1 lists all the wake-up sources.

All the wake-up sources in Table 14.4-1 can wake up both the HP CPU and the LP CPU. Users can configure [PMU_WAKEUP_ENA](#) to set the target to the HP CPU, and configure [PMU_LP_CPU_WAKEUP_EN](#) to set the target to the LP CPU. Configuring both registers simultaneously can wake up both the HP CPU and the LP CPU.

Table 14.4-1. Wake-up Sources

Bit in PMU_WAKEUP_ENA/PMU_LP_CPU_WAKEUP_EN	Wake-up Sources	Source Power Domains
0	HP_SDIO	PD_HP_CNNT
1	Software wakeup	PD_LP_PERI
2	HP GPIO	PD_TOP
3	HP USB	PD_HP_CNNT
4	HP UART4	PD_TOP
5	HP UART3	PD_TOP
6	HP UART2	PD_TOP
7	HP UART1	PD_TOP
8	HP UART0	PD_TOP
9	LP GPIO	PD_LP_PERI
10	LP UART	PD_LP_PERI
11	LP TOUCH	PD_LP_PERI
12	EXT IO	PD_LP_PERI
13	RTC_TIMER_TAR0	PD_AON
14	BROWN OUT	PD_AON
15	VBAT_UNDERVOLTAGE	PD_AON
16	LP CORE EXCEPTION	PD_LP_PERI
17	ETM	PD_TOP
18	RTC_TIMER_TAR1	PD_AON
19	LP I2S	PD_LP_PERI
20	GMAC PMT	PD_HP_CNNT
21	GMAC LPI	PD_HP_CNNT
22	LP CPU	PD_AON
23	HP CPU	PD_TOP

ESP32-P4 provides a hardware mechanism that can reject sleep, meaning if some peripherals are in an un-interruptible working state and the HP CPU tries to sleep, the peripherals will send a wake-up signal to prevent the HP CPU from sleeping, thus ensuring the peripherals work normally.

The wake-up sources in Table 14.4-1 can all be configured as events to reject sleep. Users can configure the following registers to implement sleep rejection. The configuration values of [PMU_SLEEP_REJECT_ENA](#) and [PMU_SLP_REJECT_CAUSE_REG](#) and the corresponding wake-up sources are the same as shown in Table 14.4-1.

- Enable sleep rejection feature:
 - Set [PMU_SLP_REJECT_EN](#) to 1 to enable the sleep rejection feature.
 - Configure [PMU_SLEEP_REJECT_ENA](#) to select the sleep rejection signal source.
- Read [PMU_SLP_REJECT_CAUSE_REG](#) for the source of sleep rejection event.

14.4.2.3 Analog Power Controller

The analog power controller controls the power up and down of the analog circuits (including voltage regulators, high-speed clocks, and slow-speed clocks) in the PMU states.

The configuration of the regulators is as follows:

- Configure `PMU_n1_HP_REGULATOR_XPD` or `PMU_n1_LP_REGULATOR_XPD` to enable or disable the output voltage of the HP/LP sys regulator in the target PMU state. Turning off the LP sys regulator is not recommended, as it may cause the PMU itself to power down, resulting in chip malfunction.

The configuration of the high-speed clocks (XTAL_CLK and PLL clocks) is as follows:

- XTAL_CLK: Configure `PMU_HP_SLEEP_XPD_XTAL` to 1 to enable XTAL_CLK when the chip switches the PMU state to HP_SLEEP.

Note: To avoid the instability in XTAL_CLK during startup, users have the option to configure `PMU_WAIT_XTL_STABLE` to delay the gate opening for XTAL_CLK. This delay ensures that the gate opening is enabled after `PMU_WAIT_XTL_STABLE` CLK_DYN_FAST_CLK cycles following the power-up of XTAL_CLK.

- CPLL_CLK: PMU can enable CPLL_CLK in different PMU states by configuring `PMU_n1_XPD_PLL[0]` to 1. For example, configuring `PMU_HP_ACTIVE_XPD_PLL[0]` to 1 enables the CPLL_CLK clock when the chip is in HP_ACTIVE state.
- SPLL_CLK: PMU can enable SPLL_CLK in different PMU states by configuring `PMU_n1_XPD_PLL[1]` to 1. For example, configuring `PMU_HP_ACTIVE_XPD_PLL[1]` to 1 enables the SPLL_CLK clock when the chip is in HP_ACTIVE state.
- Audio PLL: PMU can enable Audio PLL in different PMU states by configuring `PMU_n1_XPD_PLL[2]` to 1. For example, configuring `PMU_HP_ACTIVE_XPD_PLL[2]` to 1 enables the Audio PLL clock when the chip is in HP_ACTIVE state.
- SDIO PLL: PMU can enable SDIO PLL in different PMU states by configuring `PMU_n1_XPD_PLL[3]` to 1. For example, configuring `PMU_HP_ACTIVE_XPD_PLL[3]` to 1 enables the SDIO PLL clock when the chip is in HP_ACTIVE state.

Note:

- Before enabling the PLL clocks please ensure that XTAL_CLK is stable.
- To avoid the instability in the PLL clocks during startup, users have the option to configure `PMU_WAIT_PLL_STABLE` to delay the gate opening for the PLL clocks. This delay ensures that the gate opening is enabled after `PMU_WAIT_PLL_STABLE` CLK_DYN_FAST_CLK cycles following the power-up of the PLL clocks.

Slow-speed clocks (RC_FAST_CLK and XTAL32K_CLK) operate with low power. The power up and down of the slow-speed clocks in HP_ACTIVE and HP_SLEEP states are controlled by `PMU_HP_SLEEP_XPD_FOSC_CLK`. In LP_SLEEP state, the power up and down of the slow-speed clocks are controlled by `PMU_LP_SLEEP_XPD_FOSC_CLK`.

14.4.2.4 Digital Power Controller

The digital power controller controls the power up and down of digital power domains in different PMU states. Unlike the analog power controller, the digital power controller does not directly control the regulator but instead controls the power switch connected to the regulator to power up and down the digital power domains.

The LP PD Peripherals domain can only be powered up and down while the PMU state switches between

HP_SLEEP and LP_SLEEP. The other digital power domains can only be powered up and down while the PMU state switches between HP_SLEEP and HP_ACTIVE.

When the chip switches between PMU states, if the power configuration of a power domain in the current PMU state does not match the power configuration it is about to switch to, then the power up-down process will be activated. Take the power up-to-down process of the CPU power domain as an example. Configure [PMU_HP_SLEEP_PD_TOP_PD_EN](#) to 1 or 0 to power down or up the CPU power domain in HP_SLEEP. PMU will perform the following configurations:

- Enable the digital isolation unit to ensure that the powered-down modules do not output unstable voltage levels to the powered-up modules. When a power domain loses power, the output of this module will be clamped to a fixed value.
- Enable reset. When the CPU power domain loses power, its global reset signal is set to a reset state, which persists for a period after the CPU power domain is re-powered. This mechanism guarantees a reset-to-release process for the CPU power domain during power-up, effectively mitigating any instability caused by power up-down transitions.

The following explains the power up and down of each digital power domain:

- L2MEM_G0-G5

Configure [PMU_n1_PD_HP_MEM_PD_EN](#) to power up or down L2MEM_G0-G5.

- HP_CNNT

Configure [PMU_n1_PD_CNNT_PD_EN](#) to power up or down HP_CNNT.

- PD_HP_CPU

Configure [PMU_n1_PD_HP_CPU_PD_EN](#) to power up or down PD_HP_CPU.

- Peripherals + ROM (PD_TOP)

Configure [PMU_n1_PD_TOP_PD_EN](#) to power or down PD_TOP.

When the Peripherals domain is powered down, the following features are configurable:

- Powering down the Peripherals domain may cause instability in GPIOs. This can be addressed by maintaining the state of the GPIOs (excluding the LP GPIOs) through PMU. For example, configuring [PMU_HP_SLEEP_HP_PAD_HOLD_ALL](#) can keep GPIOs in the same state as before Peripherals was powered down in HP_SLEEP.
- In the HP_ACTIVE state, Memory Deep-sleep mode is supported under a standard power supply. The memory cannot be read or written in this mode, but data can be retained. Configure [PMU_n1_HP_MEM_DSLP](#) to enable this mode.

- LP PD Peripherals

The LP PD Peripherals power domain remains powered up when the chip is in HP_ACTIVE state. The power state of LP PD Peripherals is configurable only in HP_SLEEP and LP_SLEEP states. The LP PD Peripherals power domain has an independent digital power switch, and its power control is not dependent on the power state of other digital power domains.

14.4.2.5 Clock Controller

The clock controller mainly controls the high-performance system clocks and LP system clocks when the chip switches between PMU states.

The high-performance system clocks include ROOT_CLK and high-performance system peripherals clocks. When the chip switches between HP_ACTIVE and HP_SLEEP states, PMU can switch, power up/down, and divide the frequency of ROOT_CLK, as well as power up/down high-performance system peripherals clocks.

- ROOT_CLK can be controlled as follows:
 - Configure PMU_*n1*_SYS_CLK_SLP_SEL to 1 to indicate that when the chip enters the corresponding PMU state, the clock source is controlled by PMU.
 - Configure PMU_*n1*_ICG_SYS_CLOCK_EN to 0 to disable ROOT_CLK in the corresponding PMU state.
 - Configure PMU_*n1*_DIG_SYS_CLK_SEL to select the clock source after the chip enters the corresponding PMU state. For details, please refer to Chapter 10 *Reset and Clock* > Table 10.2-1.
- High-performance system peripheral clocks can be controlled as follows:
 - Configure PMU_*n1*_ICG_SLP_SEL to 1 so that the clock gating in the target state is controlled by PMU. Configure this register to 0 so that the clock gating is controlled by PCR registers.
 - Configure PMU_*n1*_DIG_ICG_FUNC_EN to power up/down the function clock in the target PMU state. For detailed configuration please see Table 14.4-2.

Table 14.4-2. HP System Peripherals' Function Clocks

Bit in PMU_ <i>n1</i> _DIG_ICG_FUNC_EN	Clock
bit 0	HP_CPU0_CLK
bit 1	HP_CPU1_CLK
bit 2	HP_CPU0_CLIC_CLK
bit 3	HP_CPU1_CLIC_CLK
bit 4	MISC_CPU_CLK
bit 5	MISC_SYS_CLK
bit 6	ICM_CPU_CLK
bit 7	ICM_SYS_CLK
bit 8	ICM_MEM_CLK
bit 9	ICM_APB_CLK
bit 10	SPM_CPU_CLK
bit 11	L2MEM_MEM_CLK
bit 12	L2MEM_SYS_CLK
bit 13	L1CACHE_CPU_CLK
bit 14	L1CACHE_D_CPU_CLK
bit 15	L1CACHE_IO_CPU_CLK
bit 16	L1CACHE_I1_CPU_CLK
bit 17	L1CACHE_MEM_CLK
bit 18	L1CACHE_D_MEM_CLK
bit 19	L1CACHE_IO_MEM_CLK

PMU_ <i>n1</i> _DIG_ICG_FUNC_EN Bit	Clock
bit 20	L1CACHE_I1_MEM_CLK
bit 21	L2CACHE_MEM_CLK
bit 22	L2CACHE_SYS_CLK
bit 23	REGDMA_SYS_CLK
bit 24	HP_CLKRST_APB_CLK
bit 25	SYSREG_APB_CLK
bit 26	INTRMTX_CLK
bit 27	N/A
bit 28	N/A
bit 29	N/A
bit 30	N/A
bit 31	N/A

The LP system clocks are mainly used in the low-power system and include the following clocks:

- LP_SLOW_CLK
- LP_FAST_CLK
- LP_DYN_SLOW_CLK
- LP_DYN_FAST_CLK
- XTAL_D2_CLK

The clock frequency of LP_DYN_FAST_CLK is controlled by hardware as follows, depending on the PMU state (and cannot be changed by the user):

- LP_SLEEP: The LP_DYN_FAST_CLK frequency is the same as LP_SLOW_CLK.
- HP_ACTIVE, HP_SLEEP: The LP_DYN_FAST_CLK frequency is the same as LP_FAST_CLK.

14.4.2.6 Backup Controller

ESP32-P4 has a Retention DMA module that can transfer data between memory and peripherals when the chip switches between PMU states. In this way, the data is backed up when the power domain is powered down and restored when the power domain is powered up again.

Data transfer is implemented in the Peripherals power domain. PMU only generates relevant control signals. It is important to note that the data transfer control registers are directional, as unlike other control registers, these control behaviors are determined by both the original PMU state and the target PMU state.

For example, the control registers for transitioning from HP_ACTIVE to HP_SLEEP and from HP_SLEEP to HP_ACTIVE are different. The possible PMU state switches are listed below, collectively represented by *n2* in the register names:

- HP_SLEEP2ACTIVE
- HP_ACTIVE2SLEEP

The following section introduces how PMU controls the Retention DMA:

- **Enable data transfer:** Configure `PMU_n2_BACKUP_EN` to 1 to enable data transfer when the corresponding PMU state switch is performed.
- **Enable corresponding clocks:** Before data transfer starts, configure `PMU_n2_BACKUP_CLK_SEL` to select the clock source of the Retention DMA, and configure `PMU_n1_BACKUP_ICG_FUNC_EN` to enable the clock.

After the data transfer is completed, the value of `PMU_n1_BACKUP_ICG_FUNC_EN` is determined by the configuration of `PMU_n1_DIG_ICG_FUNC_EN` in the target PMU state.

- **Configure data transfer direction:** Configure the highest bit of `PMU_n2_BACKUP_MODE`:
 - 1: From peripheral to memory
 - 0: From memory to peripheral
- **Select linked list pointer:** Configure the lower two bits of `PMU_n2_BACKUP_MODE` to select the linked list pointer, specifically:
 - 0: `PAU_LINK_ADDR_0`
 - 1: `PAU_LINK_ADDR_1`
 - 2: `PAU_LINK_ADDR_2`
 - 3: `PAU_LINK_ADDR_3`

14.4.2.7 System Controller

The system controller controls some functional modules when the chip switches PMU states, to achieve stable and low-power chip performance. Specifically, the system controller supports:

- **Pausing the watchdog function:** Configuring `PMU_n1_DIG_PAUSE_WDT` to 1 disables the RTC watchdog timer (RWDT) function when the chip switches to the corresponding target PMU state. Note that if this register is configured to 0 in any sleep mode, the watchdog function is not disabled, and RWDT will reset as the CPU does not feed the watchdog.
- **Switching GPIO to sleep mode,** where the configuration of the GPIO holds. Consider a GPIO pin working in low-drive mode as an input and a wake-up pin. Setting `PMU_n1_HP_PAD_HOLD_ALL` to 1 latches the current configuration of the GPIO pin as the sleep configuration when the chip transitions to the corresponding PMU state. For more information on the hold function of GPIO, please refer to Chapter 9 [GPIO Matrix and IO MUX](#) > Section 9.9.
- **Disabling UART wake-up function:** Configuring `PMU_n1_UART_WAKEUP_EN` to 0 disables the four UART wake-up modes in the corresponding PMU state. For more information on UART wake-up modes, please refer to Chapter 42 [UART Controller \(UART\)](#).
- **Pausing CPU:** Setting `PMU_n1_DIG_CPU_STALL` to 1 suspends the CPU in the corresponding PMU state.

14.4.2.8 Battery Power Controller

As shown in Figure 14.4-1, the low-power system regulator can be powered by `VDD_BAT` (battery) or `VDD_ANA` (analog). `PMU_n1_VDDBAT_MODE` controls whether to enable the battery power. When `PMU_n1_VDDBAT_MODE` is 0, the battery power is not enabled, and the low-power system regulator is powered by `VDD_ANA` and can be used in Active mode or Light-sleep0/1 modes. When

`PMU_n1_VDDBAT_MODE` is 1, the battery power is enabled and can be used in Deep-sleep mode, where the main power, including VDD_ANA, is cut off, and only the LP AON & PMU power domains are powered up.

14.4.2.9 Output Regulator Control

As shown in Figure 14.4-1, ESP32-P4 has four output regulators:

- VDDO_FLASH (VO1), typically used for on-chip flash voltage regulation
- VDDO_PSRAM (VO2), typically used for on-chip PSRAM voltage regulation
- VDDO_3 (VO3) and VDDO_4 (VO4), which convert the input VDD_LDO (typically 3.3 V) into the voltages needed by peripherals such as flash, PSRAM, MIPI, and SDIO.

14.4.2.9.1 Regulator Features

The regulators support the following features:

- **Supports LDO mode and bypass mode.** LDO mode supports an output of 0.5 V – 2.7 V, while bypass mode outputs the voltage of VDD_LDO. Mode selection is controlled by a hardware state machine or a register, with the control authority managed by a register. The flash regulator (VO1) also supports eFuse control.
 - The mode control for VO1 is shown in Figure 14.4-3. In the figure, TIEH_SEL on the right being 0 selects LDO mode; being 1 selects bypass mode.

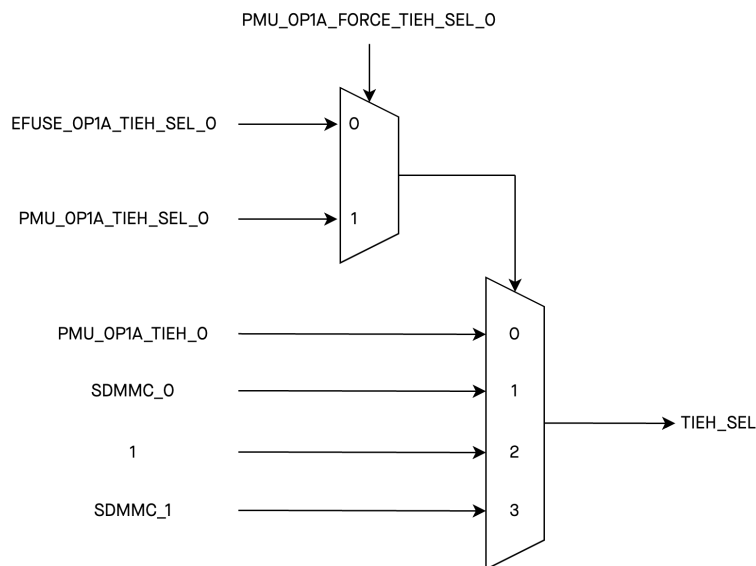


Figure 14.4-3. VO1 Mode Control

- The mode control for VO2/3/4 is shown in Figure 14.4-4. In the figure, TIEH_SEL on the right being 0 selects LDO mode; being 1 selects bypass mode.

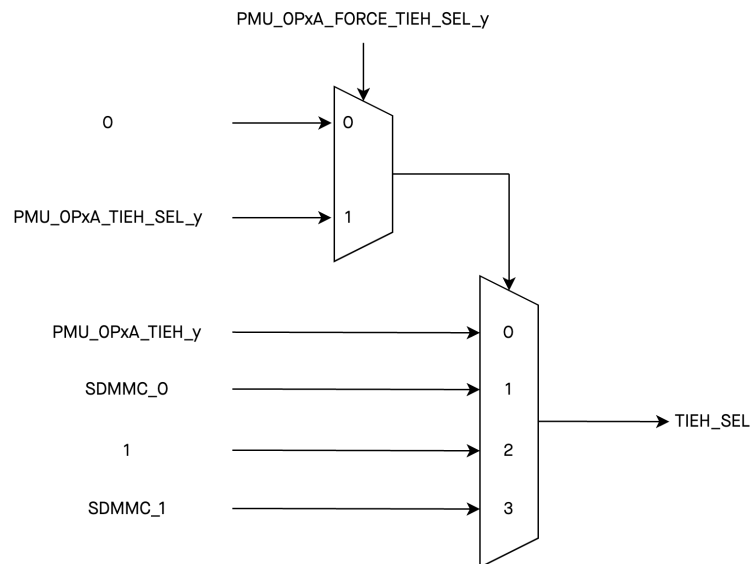


Figure 14.4-4. VO2/3/4 Mode Control

Table 14.4-3. VO2/3/4 Registers

Regulators	PMU_OPxA_FORCE_TIEH_SEL_y	PMU_OPxA_TIEH_SEL_y	PMU_OPxA_TIEH_y
VO2	PMU_OP1A_FORCE_TIEH_SEL_1	PMU_OP1A_TIEH_SEL_1	PMU_OP1A_TIEH_1
VO3	PMU_OP2A_FORCE_TIEH_SEL_0	PMU_OP2A_TIEH_SEL_0	PMU_OP2A_TIEH_0
VO4	PMU_OP2A_FORCE_TIEH_SEL_1	PMU_OP2A_TIEH_SEL_1	PMU_OP2A_TIEH_1

- Supports LDO output voltage setting:** In LDO mode, the output voltage is configured by analog registers `PMU_ANA_0PxA_DREF_y` (sets V_{ref}) and `PMU_ANA_0PxA_MUL_y` (sets MUL), following $V_{OUT} = MUL \times V_{REF}$. V_{ref} supports 0.5 V–1.6 V, and MUL supports 1.0–2.75 (step 0.25). For details, see field descriptions in [PMU_EXT_LDO_PO_OP1A_ANA_REG](#)–[PMU_EXT_LDO_PO_OP2A_ANA_REG](#).
- Supports switch control.** Only the flash voltage regulator (VO1) supports power control by the PMU state machine, the other regulators only support power control by software. The regulator is enabled when the register in Table 14.4-4 is set to 1; otherwise, the regulator is disabled.

Table 14.4-4. Regulator Power Control

Regulator	Power Control
VO1	<code>PMU_n1_REGULATOR_VO1_XPD</code>
VO2	PMU_OP1A_XPD_1
VO3	PMU_OP2A_XPD_0
VO4	PMU_OP2A_XPD_1

- Over-current protection:** Before powering up the regulators, users have the option to enable the over-current protection function to prevent excessive transient current that may damage the capacitor. After the capacitor charging is complete, the over-current protection should be disabled. The capacitor charging time depends on the load capacitance and voltage, following $T_{ms} = C_F \times V_V / 50$. Table 14.4-5 provides the over-current protection enable registers; when the register is set to 1, the function is enabled; otherwise, it is disabled.

Table 14.4-5. Over-Current Protection Control

Regulators	Over-Current Protection Control Registers
VO1	FSM or PMU_ANA_OP1A_EN_CUR_LIM_0
VO2	PMU_ANA_OP1A_EN_CUR_LIM_1
VO3	PMU_ANA_OP2A_EN_CUR_LIM_0
VO4	PMU_ANA_OP2A_EN_CUR_LIM_1

Due to the long power-up time, a waiting counter is provided for each regulator. Once the regulator is enabled, the counter is triggered and starts counting. The counter has two stages (stage 1/stage 2, corresponding to registers **TARGET0/TARGET1**), and each stage triggers different interrupts. Typically, set stage 1 (**TARGET0**) to over-current protection duration, and stage 2 (**TARGET1**) to voltage stabilization waiting time. The configuration of the counter is shown in the table below:

Table 14.4-6. Regulator Waiting Counter

Regulators	Counter Target 1	Counter Target 2
VO1	PMU_OP1A_TARGET0_0[6:0]	PMU_OP1A_TARGET1_0[7:0]
VO2	PMU_OP1A_TARGET0_1[7:0]	PMU_OP1A_TARGET1_1[7:0]
VO3	PMU_OP2A_TARGET0_0[7:0]	PMU_OP2A_TARGET1_0[7:0]
VO4	PMU_OP2A_TARGET0_1[7:0]	PMU_OP2A_TARGET1_1[7:0]

Note:

1. The bit 7 of [PMU_OP1A_TARGET0_0](#) clears the counter. Setting bit 7 to 1 resets the counter.
2. The counter clock is derived from LP_DYN_SLOW_CLK, which is divided by a factor of 16.

14.4.2.9.2 Regulator Configuration Examples**– Example 1: VO3 powering MIPI D-PHY (2.5 V, LDO mode)**

On the board, the VO3 pin is connected to VDD_MIPIDPHY that requires 2.5 V operating voltage. In this scenario, select LDO mode and set $V_{OUT} = 2.5\text{ V}$. One valid combination is $V_{REF} = 1.0\text{ V}$ (DREF=9) and MUL = 2.5 (MUL=6), i.e., [PMU_ANA_OP2A_DREF_0](#)=9, [PMU_ANA_OP2A_MUL_0](#)=6, and set [PMU_OP2A_TIEH_0](#)=0 (LDO).

– Example 2: VO4 powering VDD_IO_5 (SD card voltage switching)

In reference designs, VO4 may be connected to VDD_IO_5 to supply some GPIO/SDIO pins.

- 3.3 V output: use bypass mode (for example, set [PMU_OP2A_TIEH_1](#)=1).
- 1.8 V output: use LDO mode and set $V_{OUT} = 1.8\text{ V}$. One valid combination is $V_{REF} = 0.9\text{ V}$ (DREF=8) and MUL = 2.0 (MUL=4), i.e., [PMU_ANA_OP2A_DREF_1](#)=8, [PMU_ANA_OP2A_MUL_1](#)=4, and set [PMU_OP2A_TIEH_1](#)=0 (LDO).

14.4.2.9.3 Software Configuration Steps

The recommended software configuration steps (example) are as follows:

1. Configure output mode: set [PMU_OPxA_FORCE_TIEH_SEL_y](#) so the register controls mode selection, then choose LDO or bypass mode via [PMU_OPxA_TIEH_SEL_y](#)/[PMU_OPxA_TIEH_y](#) (see Table 14.4-3).

2. If using LDO mode, configure output voltage: configure `PMU_ANA_0PxA_DREF_y` and `PMU_ANA_0PxA_MUL_y` to satisfy $V_{OUT} = MUL \times V_{REF}$.
3. Configure waiting counter: set stage 1 (`TARGET0`) to the desired over-current protection duration and stage 2 (`TARGET1`) to the voltage stabilization waiting time (see Table 14.4-6).
4. Enable over-current protection before power-up: set `PMU_ANA_0PxA_EN_CUR_LIM_y=1` (see Table 14.4-5).
5. Enable the regulator: set `PMU_0PxA_XPD_y=1` (see Table 14.4-4), then wait for the stage 1 timeout interrupt (e.g., VO3 uses `PMU_0P2A_CNT_TARGET0_REACH_0_INT`, VO4 uses `PMU_0P2A_CNT_TARGET0_REACH_1_INT`).
6. After stage 1 interrupt, disable over-current protection: clear `PMU_ANA_0PxA_EN_CUR_LIM_y`. If a voltage stabilization waiting time is required, wait for the stage 2 timeout interrupt (`...CNT_TARGET1_REACH...INT`).

14.5 Power Modes

ESP32-P4 has three configurable PMU states. Based on the PMU states, the chip has defined five power modes for common application scenarios, as shown in Table 14.5-1.

Table 14.5-1. Preset Power Modes

Power Modes	Power Domains							
	LP always-on	LP PD peripherals	HP_CPU	TOP	HPCNNT	RC_FAST_CLK	XTAL_CLK	PLL
Active	ON	ON	ON	ON	ON	ON	ON	ON
Light-sleep0	ON	ON	OFF	ON	ON	ON	ON	ON
Light-sleep1	ON	ON	OFF	ON	ON	ON/OFF	ON/OFF	ON/OFF
Light-sleep2	ON	ON/OFF	OFF	OFF	ON	ON/OFF	ON/OFF	ON/OFF
Deep-sleep	ON	OFF	OFF	OFF	OFF	OFF	OFF	OFF

Note:

1. For power consumption data, please refer to [ESP32-P4 Datasheet](#).
2. For supported wake-up sources, please refer to Table 14.4-1.

14.6 Event Task Matrix Feature

The low-power management system on ESP32-P4 supports the Event Task Matrix (ETM) function, which allows the low-power management system's ETM tasks to be triggered by any peripherals' ETM events, or the low-power management system's ETM events to trigger any peripherals' ETM tasks. This section introduces the ETM tasks and events related to the low-power management system. For more information, please refer to Chapter 13 [Event Task Matrix \(ETM\)](#).

The low-power management system can receive the following ETM tasks:

- PMU_TASK_SLEEP_REQ: Triggers PMU's sleep process.

The low-power management system can generate the following ETM events:

- PMU_EVT_SLEEP_WAKEUP: Indicates that PMU is woken up to the HP_ACTIVE state.

14.7 Interrupts

ESP32-P4's low-power management system can generate the following interrupt signals:

- PMU_INTR
- PMU_LP_INT

Among these interrupt signals, PMU_INTR is sent to the [Interrupt Matrix](#), while PMU_LP_INT is sent to the LP CPU.

The interrupt signals are generated by the internal interrupt sources of each module, specifically:

PMU_INTR:

- PMU_SOC_WAKEUP_INT: Triggered when the chip is woken up to the HP_ACTIVE state.
- PMU_SOC_SLEEP_REJECT_INT: Triggered when a sleep-rejection source rejects a sleep request.
- PMU_SW_INT: Triggered when LP CPU is used as a wake-up source and wakes up the chip to HP_ACTIVE state.
- PMU_SDIO_IDLE_INT: Triggered when SDIO is in idle state.
- PMU_LP_CPU_EXC_INT: Triggered when there are LP CPU exceptions.
- PMU_OP2A_CNT_TARGET1_REACH_1_INT: Triggered when VO4 regulator's waiting counter target 1 reaches timeout.
- PMU_OP2A_CNT_TARGET0_REACH_1_INT: Triggered when VO4 regulator's waiting counter target 0 reaches timeout.
- PMU_OP2A_CNT_TARGET1_REACH_0_INT: Triggered when VO3 regulator's waiting counter target 1 reaches timeout.
- PMU_OP2A_CNT_TARGET0_REACH_0_INT: Triggered when VO3 regulator's waiting counter target 0 reaches timeout.
- PMU_OP1A_CNT_TARGET1_REACH_1_INT: Triggered when VO2 regulator's waiting counter target 1 reaches timeout.
- PMU_OP1A_CNT_TARGET0_REACH_1_INT: Triggered when VO2 regulator's waiting counter target 0 reaches timeout.
- PMU_OP1A_CNT_TARGET1_REACH_0_INT: Triggered when VO1 regulator's waiting counter target 1 reaches timeout.
- PMU_OP1A_CNT_TARGET0_REACH_0_INT: Triggered when VO1 regulator's waiting counter target 0 reaches timeout.

PMU_LP_INT:

- PMU_LP_CPU_SLEEP_REJECT_INT: Triggered when the LP sleep request is rejected.

- `PMU_HP_SW_TRIGGER_INT`: Triggered when HP CPU wakes up LP CPU.
- `PMU_ACTIVE_SWITCH_SLEEP_START_INT`: Triggered when the PMU state starts to switch from `HP_ACTIVE` to `HP_SLEEP`.
- `PMU_SLEEP_SWITCH_ACTIVE_START_INT`: Triggered when the PMU state starts to switch from `HP_SLEEP` to `HP_ACTIVE`.
- `PMU_ACTIVE_SWITCH_SLEEP_END_INT`: Triggered when the PMU state finishes switching from `HP_ACTIVE` to `HP_SLEEP`.
- `PMU_SLEEP_SWITCH_ACTIVE_END_INT`: Triggered when the PMU state finishes switching from `HP_SLEEP` to `HP_ACTIVE`.
- `PMU_LP_CPU_WAKEUP_INT`: Triggered when LP CPU is woken up.
- `PMU_OP2A_CNT_TARGET1_REACH_1_LP_INT`: Triggered when VO4 regulator's waiting counter target 1 reaches timeout.
- `PMU_OP2A_CNT_TARGET0_REACH_1_LP_INT`: Triggered when VO4 regulator's waiting counter target 0 reaches timeout.
- `PMU_OP2A_CNT_TARGET1_REACH_0_LP_INT`: Triggered when VO3 regulator's waiting counter target 1 reaches timeout.
- `PMU_OP2A_CNT_TARGET0_REACH_0_LP_INT`: Triggered when VO3 regulator's waiting counter target 0 reaches timeout.
- `PMU_OP1A_CNT_TARGET1_REACH_1_LP_INT`: Triggered when VO2 regulator's waiting counter target 1 reaches timeout.
- `PMU_OP1A_CNT_TARGET0_REACH_1_LP_INT`: Triggered when VO2 regulator's waiting counter target 0 reaches timeout.
- `PMU_OP1A_CNT_TARGET1_REACH_0_LP_INT`: Triggered when VO1 regulator's waiting counter target 1 reaches timeout.
- `PMU_OP1A_CNT_TARGET0_REACH_0_LP_INT`: Triggered when VO1 regulator's waiting counter target 0 reaches timeout.

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [14.8 Register Summary](#).

14.8 Register Summary

The addresses in this section are relative to the PMU base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
Configuration Registers			
PMU_HP_ACTIVE_DIG_POWER_REG	Digital power domain control register in HP_ACTIVE state	0x0000	R/W
PMU_HP_ACTIVE_ICG_HP_FUNC_REG	HP system peripheral's function clock control register in HP_ACTIVE state	0x0004	R/W
PMU_HP_ACTIVE_HP_SYS_CNTL_REG	System control register in HP_ACTIVE state	0x0010	R/W
PMU_HP_ACTIVE_HP_CK_POWER_REG	Clock source power control register in HP_ACTIVE state	0x0014	R/W
PMU_HP_ACTIVE_BIAS_REG	BIAS control register in HP_ACTIVE state	0x0018	R/W
PMU_HP_ACTIVE_BACKUP_REG	Backup module control register in HP_ACTIVE state	0x001C	R/W
PMU_HP_ACTIVE_BACKUP_CLK_REG	Backup module's function clock control register in HP_ACTIVE state	0x0020	R/W
PMU_HP_ACTIVE_SYSCLK_REG	System clock control register in HP_ACTIVE state	0x0024	R/W
PMU_HP_ACTIVE_HP_REGULATORO_REG	Regulator power control register in HP_ACTIVE state	0x0028	varies
PMU_HP_ACTIVE_XTAL_REG	XTAL_CLK power control register in HP_ACTIVE state	0x0030	R/W
PMU_HP_SLEEP_DIG_POWER_REG	Digital power domain control register in HP_SLEEP state	0x0068	R/W
PMU_HP_SLEEP_ICG_HP_FUNC_REG	HP system peripheral's function clock control register in HP_SLEEP state	0x006C	R/W
PMU_HP_SLEEP_HP_SYS_CNTL_REG	System control register in HP_SLEEP state	0x0078	R/W
PMU_HP_SLEEP_HP_CK_POWER_REG	Clock source power control register in HP_SLEEP state	0x007C	R/W
PMU_HP_SLEEP_BIAS_REG	BIAS control register in HP_SLEEP state	0x0080	R/W
PMU_HP_SLEEP_BACKUP_REG	Backup module control register in HP_SLEEP state	0x0084	R/W
PMU_HP_SLEEP_BACKUP_CLK_REG	Backup module's function clock control register in HP_SLEEP state	0x0088	R/W
PMU_HP_SLEEP_SYSCLK_REG	System clock control register in HP_SLEEP state	0x008C	R/W
PMU_HP_SLEEP_HP_REGULATORO_REG	Regulator power control register in HP_SLEEP state	0x0090	R/W

Name	Description	Address	Access
PMU_HP_SLEEP_XTAL_REG	XTAL_CLK power control register in HP_SLEEP state	0x0098	R/W
PMU_HP_SLEEP_LP_REGULATOR0_REG	LP sys regulator control register in HP_SLEEP state	0x009C	R/W
PMU_HP_SLEEP_LP_DIG_POWER_REG	LP system digital power domains control register in HP_SLEEP state	0x00A8	R/W
PMU_HP_SLEEP_LP_CK_POWER_REG	Low-speed clock power control register in HP_ACTIVE/HP_SLEEP state	0x00AC	R/W
PMU_LP_SLEEP_LP_REGULATOR0_REG	LP sys regulator control register in HP_SLEEP state	0x00B4	R/W
PMU_LP_SLEEP_XTAL_REG	XTAL_CLK power control register in LP_SLEEP state	0x00BC	R/W
PMU_LP_SLEEP_LP_DIG_POWER_REG	Digital power domain control register in LP_SLEEP state	0x00C0	R/W
PMU_LP_SLEEP_LP_CK_POWER_REG	Low-speed clock power control register in LP_SLEEP state	0x00C4	R/W
PMU_IMM_HP_CK_POWER_REG	Software control register for clock power	0x00CC	WT
PMU_IMM_SLEEP_SYSCLK_REG	Software control register for system clock	0x00D0	WT
PMU_IMM_HP_FUNC_ICG_REG	Software control register for peripheral clock	0x00D4	WT
PMU_IMM_HP_APB_ICG_REG	Software control register for APB clock	0x00D8	WT
PMU_IMM_PAD_HOLD_ALL_REG	Software control register for PAD hold function	0x00E4	varies
PMU_POWER_CK_WAIT_CNTL_REG	Clock gating wait time configuration register	0x011C	R/W
PMU_SLP_WAKEUP_CNTL0_REG	Sleep request register	0x0120	WT
PMU_SLP_WAKEUP_CNTL1_REG	Sleep reject register	0x0124	R/W
PMU_SLP_WAKEUP_CNTL2_REG	Wake up source enable register	0x0128	R/W
PMU_SLP_WAKEUP_CNTL3_REG	Minimum sleep time control register	0x012C	R/W
PMU_SLP_WAKEUP_CNTL4_REG	Sleep reject cause clear register	0x0130	WT
PMU_SLP_WAKEUP_STATUS0_REG	Wake up cause register	0x0144	RO
PMU_SLP_WAKEUP_STATUS1_REG	Reset reject cause register	0x0148	RO
PMU_HP_CK_CNTL_REG	HP system clock control register	0x0154	R/W
PMU_RF_PWC_REG	SAR ADC power up register	0x015C	R/W
PMU_INT_RAW_REG	PMU sleep/wake-up raw interrupt	0x0164	R/WTC/SS
PMU_HP_INT_ST_REG	PMU sleep/wake-up state interrupt	0x0168	RO
PMU_HP_INT_ENA_REG	PMU sleep/wake-up interrupt enable register	0x016C	R/W
PMU_HP_INT_CLR_REG	PMU sleep/wake-up interrupt clear register	0x0170	WT
PMU_LP_INT_RAW_REG	PMU sleep/wake-up raw interrupt	0x0174	R/WTC/SS
PMU_LP_INT_ST_REG	PMU sleep/wake-up state interrupt	0x0178	RO

Name	Description	Address	Access
PMU_LP_INT_ENA_REG	PMU sleep/wake-up interrupt enable register	0x017C	R/W
PMU_LP_INT_CLR_REG	PMU sleep/wake-up interrupt clear register	0x0180	varies
PMU_LP_CPU_PWR0_REG	LP CPU sleep/wake-up control register	0x0184	varies
PMU_LP_CPU_PWR1_REG	LP CPU sleep/wake-up control register	0x0188	WT
PMU_LP_CPU_PWR2_REG	LP CPU sleep/wake-up control register	0x018C	R/W
PMU_LP_CPU_PWR3_REG	LP CPU sleep/wake-up control register	0x0190	RO
PMU_LP_CPU_PWR4_REG	LP CPU sleep/wake-up control register	0x0194	R/W
PMU_LP_CPU_PWR5_REG	LP CPU sleep/wake-up control register	0x0198	RO
PMU_HP_LP_CPU_COMM_REG	LP CPU sleep/wake-up control register	0x019C	WT
PMU_EXT_LDO_PO_OP1A_REG	VO1 regulator control register	0x01B8	R/W
PMU_EXT_LDO_PO_OP1A_ANA_REG	VO1 regulator control register	0x01BC	R/W
PMU_EXT_LDO_PO_OP2A_REG	VO3 regulator control register	0x01C0	R/W
PMU_EXT_LDO_PO_OP2A_ANA_REG	VO3 regulator control register	0x01C4	R/W
PMU_EXT_LDO_P1_OP1A_REG	VO2 regulator control register	0x01D0	R/W
PMU_EXT_LDO_P1_OP1A_ANA_REG	VO2 regulator control register	0x01D4	R/W
PMU_EXT_LDO_P1_OP2A_REG	VO4 regulator control register	0x01D8	R/W
PMU_EXT_LDO_P1_OP2A_ANA_REG	VO4 regulator control register	0x01DC	R/W
PMU_EXT_WAKEUP_LV_REG	EXT wake-up level control register	0x01E8	R/W
PMU_EXT_WAKEUP_SEL_REG	EXT wake-up IO control register	0x01EC	R/W
PMU_EXT_WAKEUP_ST_REG	EXT wake-up status register	0x01F0	RO
PMU_EXT_WAKEUP_CNTL_REG	EXT wake-up control register	0x01F4	R/W
PMU_SDIO_WAKEUP_CNTL_REG	SDIO wake-up control register	0x01F8	R/W
PMU_TOUCH_PWR_CNTL_REG	TOUCH sleep/wake-up control register	0x0210	R/W
PMU_DATE_REG	Version control register	0x03FC	R/W

14.9 Registers

The addresses in this section are relative to the PMU base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 14.1. PMU_HP_ACTIVE_DIG_POWER_REG (0x0000)

[illegible]

PMU_HP_ACTIVE_DCDC_SWITCH_PD_EN	Configures whether to cut off the power from external DCDC in HP ACTIVE state.
--	--

0: Do not cut off

1: Cut off

(R/W)

PMU_HP_ACTIVE_HP_MEM_DSLP Configures whether to put L2MEM_G0-G5 into Deep-sleep mode in HP_ACTIVE state.

0: Do not put L2MEM_G0-G5 into Deep-sleep

1: Put L2MEM_G0-G5 into Deep-sleep

(R/W)

PMU_HP_ACTIVE_PD_HP_MEM_PD_EN Configures whether to power down L2MEM_G0-G5 in HP ACTIVE state.

0: Power up

1: Power down

(R/W)

PMU_HP_ACTIVE_PD_HP_CPU_PD_EN	Configures whether to power down the HP_CPU power domain in HP_ACTIVE state.
--------------------------------------	--

0: Power up

1: Power down

(R/W)

PMU_HP_ACTIVE_PD_CNNT_PD_EN Configures whether to power down HP_CNNT in HP_ACTIVE state.

0: Power up

1: Power down

(R/W)

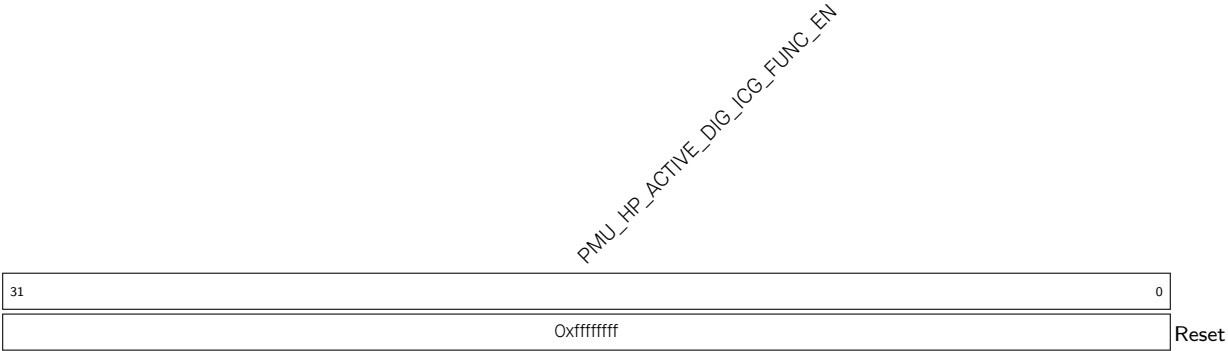
PMU_HP_ACTIVE_PD_TOP_PD_EN Configures whether to power down PD_TOP power domain in HP ACTIVE state.

0: Power up

1: Power down

(R/W)

Register 14.2. PMU_HP_ACTIVE_ICG_HP_FUNC_REG (0x0004)



PMU_HP_ACTIVE_DIG_ICG_FUNC_EN Configures whether to enable HP system peripheral's function clock in HP_ACTIVE state. For detailed configuration please see Table 14.4-2.

0: Disable

1: Enable

(R/W)

Register 14.3. PMU_HP_ACTIVE_HP_SYS_CNTL_REG (0x0010)

Diagram illustrating the PMU_CTL register structure. The register is 32 bits wide, with bits 31 down to 23 labeled as (reserved). Bits 22 down to 0 are labeled with various PMU control fields: PMU_HP_ACTIVE_DIG_CPU_STALL, PMU_HP_ACTIVE_DIG_PAUSE_WDT, PMU_HP_ACTIVE_DIG_PAD_SLP_SEL, PMU_HP_ACTIVE_LP_PAD_HOLD_ALL, and PMU_HP_ACTIVE_UART_WAKEUP_EN. A 'Reset' label is at the bottom right.

PMU_HP_ACTIVE_UART_WAKEUP_EN Configures whether to enable UART wake up function in HP ACTIVE state.

0: Disable wake-up function

1: Enable wake-up function

(R/W)

PMU_HP_ACTIVE_LP_PAD_HOLD_ALL Configures whether to hold LP GPIO's configuration in HP ACTIVE state.

0: Do not hold

1: Hold

(R/W)

PMU_HP_ACTIVE_HP_PAD_HOLD_ALL Configures whether to hold GPIO's configuration in HP ACTIVE state.

0: Do not hold

1: Hold

(R/W)

PMU_HP_ACTIVE_DIG_PAD_SLP_SEL Configures whether to use Light-sleep mode configuration for GPIO in HP_ACTIVE state.

0: Use normal configuration

1: Use Light-sleep mode configuration. For details please refer to Chapter 9 *GPIO Matrix and IO MUX* > Section 9.8 *Pin Functions in Light-sleep*.

(R/W)

PMU_HP_ACTIVE_DIG_PAUSE_WDT Configures whether to pause watchdog in HP_ACTIVE state.

0: Do not pause

1: Pause

(R/W)

PMU_HP_ACTIVE_DIG_CPU_STALL Configures whether to stall the CPU in HP_ACTIVE state.

0: Do not stall

1: Stall

(R/W)

Register 14.4. PMU_HP_ACTIVE_HP_CK_POWER_REG (0x0014)

[illegible]

PMU_HP_ACTIVE_I2C_ISO_EN Configure whether to power up ANALOG_I2C in HP_ACTIVE state.

0: Power down

1: Power up

(R/W)

PMU_HP_ACTIVE_I2C_RETENTION Configures whether to enable ANALOG_I2C retention status in HP ACTIVE state.

0: Disable

1: Enable

(R/W)

PMU_HP_ACTIVE_XPD_PLL_I2C Configures whether to enable PLL I2C controllers in HP_ACTIVE state. Each bit controls a PLL I2C register group.

0: Disable

1: Enable

(R/W)

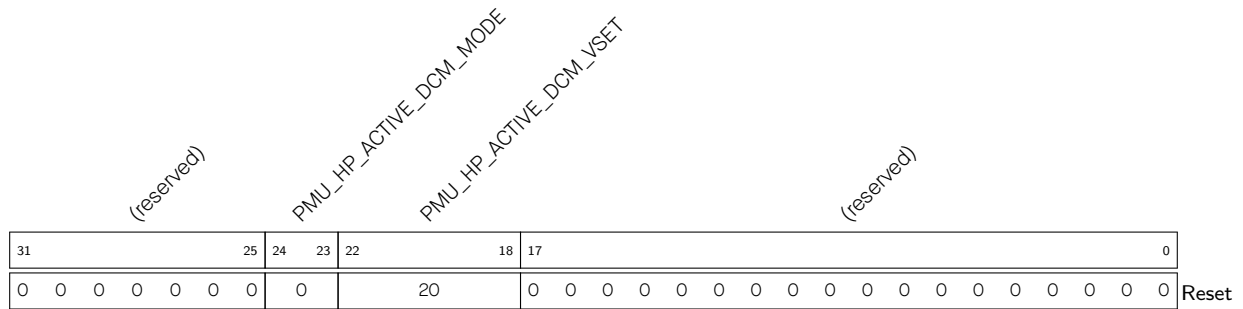
PMU_HP_ACTIVE_XPD_PLL Configures whether to enable the analog power domains in HP system in HP_ACTIVE state. Bit 0 controls CPLL_CLK, bit 1 controls SPPL_CLK, bit 2 controls Audio PLL, bit 3 controls SDIO PLL.

0: Disable

1: Enable

(R/W)

Register 14.5. PMU_HP_ACTIVE_BIAS_REG (0x0018)



PMU_HP_ACTIVE_DCM_VSET Regulates the DCDC voltage in HP_ACTIVE state. (R/W)

PMU_HP_ACTIVE_DCM_MODE Configures whether to enable DCDC status in HP_ACTIVE state.

0: Disable

1: Enable

2, 3: Reserved

(R/W)

Register 14.6. PMU_HP_ACTIVE_BACKUP_REG (0x001C)

[illegible]

PMU_HP_SLEEP2ACTIVE_BACKUP_CLK_SEL Configures the backup module's function clock source when PMU state switches from HP_SLEEP to HP_ACTIVE.

0: Select XTAL

1: Select PLL CLK

2: Select RC FAST CLK

3: Invalid value

(R/W)

PMU_HP_SLEEP2ACTIVE_BACKUP_MODE Configures the backup direction and link list when PMU state switches switch from HP_SLEEP to HP_ACTIVE.

Highest bit:

0: From peripheral to memory

1: From memory to peripheral

Lower two bits:

0: PAU_LINK_ADDR_0

1: PAU_LINK_ADDR_1

2: PAU_LINK_ADDR_2

3: PAU_LINK_ADDR_3

(R/W)

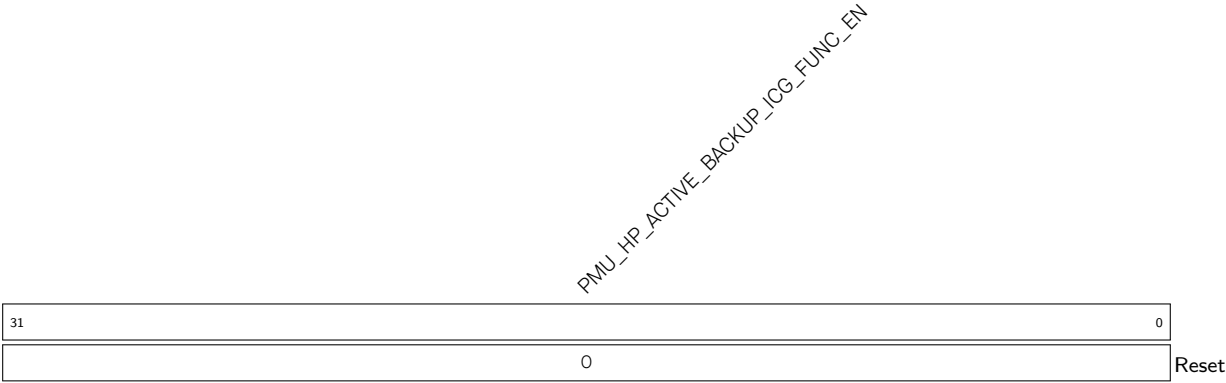
PMU_HP_SLEEP2ACTIVE_BACKUP_EN Configures whether to enable the backup flow when PMU state switches from HP_SLEEP to HP_ACTIVE.

0: Disable backup

1: Enable backup

(R/W)

Register 14.7. PMU_HP_ACTIVE_BACKUP_CLK_REG (0x0020)



PMU_HP_ACTIVE_BACKUP_ICG_FUNC_EN Configures whether to enable each peripheral's function clock when the target state is HP_ACTIVE. For details, please refer to Table 14.4-2.

0: Disable

1: Enable

(R/W)

Register 14.8. PMU_HP_ACTIVE_SYSCLK_REG (0x0024)

[illegible]

PMU_HP_ACTIVE_ICG_SYS_CLOCK_EN Configures whether to enable ROOT_CLK in HP_ACTIVE state.

0: Disable

1: Enable

(R/W)

PMU_HP_ACTIVE_SYS_CLK_SLP_SEL Configures whether to allow PMU to control clock source in HP ACTIVE state.

0: Controlled by PCR registers

1: Controlled by PMU

(R/W)

PMU_HP_ACTIVE_ICG_SLP_SEL Configures whether to allow PMU to control the clock gating in HP ACTIVE state.

0: Controlled by PCR registers

1: Controlled by PMU

(R/W)

PMU_HP_ACTIVE_DIG_SYS_CLK_SEL Configures the source of ROOT_CLK in HP_ACTIVE state.

O: XTAL

1: PLL CLK

2: RC_FAST_CLK

3: Invalid value

(R/W)

Register 14.9. PMU_HP_ACTIVE_HP_REGULATOR0_REG (0x0028)

PMU_HP_ACTIVE_HP_REGULATOR_DBIAS				(reserved)				PMU_HP_ACTIVE_REGULATOR_VO1_XPD				(reserved)				PMU_HP_ACTIVE_HP_REGULATOR_XPD				(reserved)				PMU_DIG_DBIAS_INIT				PMU_DIG_REGULATOR0_DBIAS_SEL				PMU_HP_DBIAS_VOL				PMU_LP_DBIAS_VOL				(reserved)			
31	27	26	23	22	21	19	18	17	16	15	14	13	9	8	4	3	0																										
24				0	0	0	0	1	0	0	0	1	0	0	0	1	24	24	0	0	0	0	Reset																				

PMU_LP_DBIAS_VOL Indicates the current voltage of the LP system regulator. (RO)

PMU_HP_DBIAS_VOL Indicates the current voltage of the HP system regulator. (RO)

PMU_DIG_REGULATOR0_DBIAS_SEL Configures which of the following regulates the HP system regulator voltage.

0: Regulated by Hardware automatically

1: Regulated by Software

(R/W)

PMU_DIG_DBIAS_INIT Initializes the PVT voltage configurations. (WT)

PMU_HP_ACTIVE_HP_REGULATOR_XPD Configures whether to enable the HP sys regulator in HP_ACTIVE state.

0: Disable the HP sys regulator

1: Enable the HP sys regulator

(R/W)

PMU_HP_ACTIVE_REGULATOR_VO1_XPD Configures whether to enable VO1 regulator in HP_ACTIVE state.

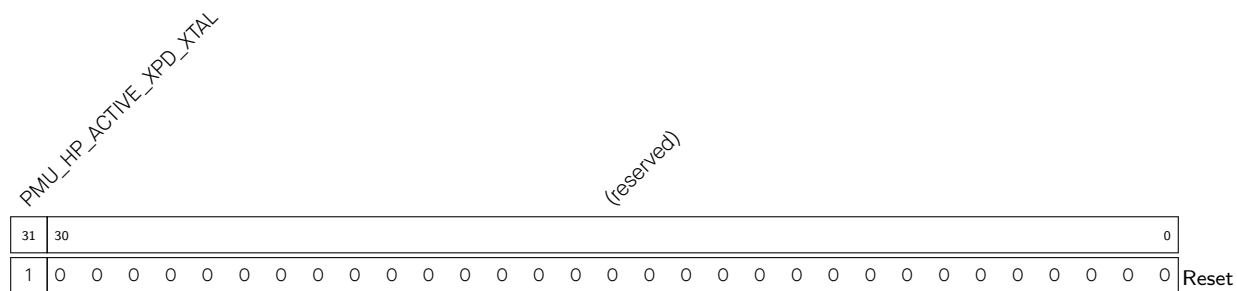
0: Disable

1: Enable

(R/W)

PMU_HP_ACTIVE_HP_REGULATOR_DBIAS Regulates the voltage of the HP sys regulator in HP_ACTIVE state. The higher the value, the higher the voltage. (R/W)

Register 14.10. PMU_HP_ACTIVE_XTAL_REG (0x0030)



PMU_HP_ACTIVE_XPD_XTAL Configures whether to enable XTAL_CLK analog source in HP_ACTIVE state.

0: Disable

1: Enable

(R/W)

Register 14.11. PMU_HP_SLEEP_DIG_POWER_REG (0x0068)

[illegible]

PMU_HP_SLEEP_DCDC_SWITCH_PD_EN Configures whether to cut off the power from external DCDC in HP SLEEP state.

0: Do not cut off

1: Cut off

(R/W)

PMU_HP_SLEEP_HP_MEM_DSLP Configures whether to put L2MEM_G0-G5 into Deep-sleep mode in HP_SLEEP state.

0: Do not put L2MEM_G0-G5 into Deep-sleep

1: Put L2MEM_G0-G5 into Deep-sleep

(R/W)

PMU_HP_SLEEP_PD_HP_MEM_PD_EN Configures whether to power down L2MEM_G0-G5 in HP SLEEP state.

0: Power up

1: Power down

(R/W)

PMU_HP_SLEEP_PD_HP_CPU_PD_EN	Configures whether to power down the HP_CPU power domain in HP_SLEEP state.
-------------------------------------	---

0: Power up

1: Power down

(R/W)

PMU_HP_SLEEP_PD_CNNT_PD_EN Configures whether to power down HP_CNNT in HP_SLEEP state.

0: Power up

1: Power down

(R/W)

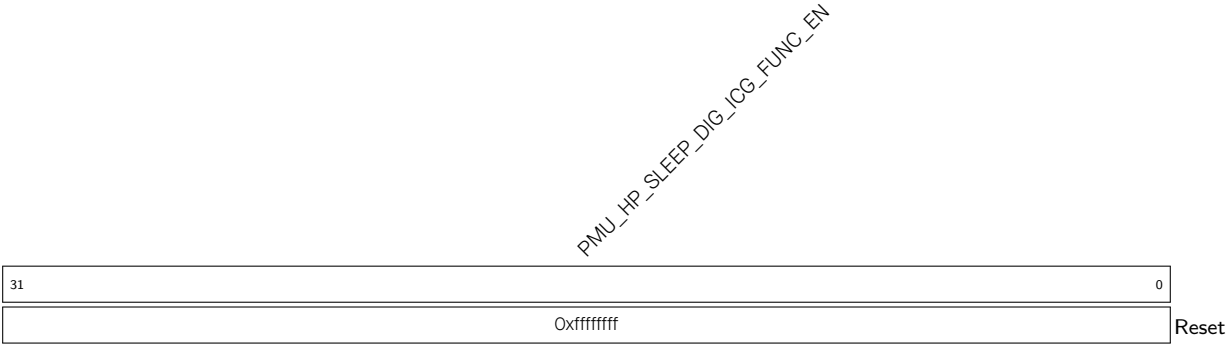
PMU_HP_SLEEP_PD_TOP_PD_EN Configures whether to power down PD_TOP power domain in HP_SLEEP state.

0: Power up

1: Power down

(R/W)

Register 14.12. PMU_HP_SLEEP_ICG_HP_FUNC_REG (0x006C)



PMU_HP_SLEEP_DIG_ICG_FUNC_EN Configures whether to enable HP system peripheral's function clock in HP_SLEEP state. For detailed configuration please see Table 14.4-2.

0: Disable

1: Enable

(R/W)

Register 14.13. PMU_HP_SLEEP_HP_SYS_CNTL_REG (0x0078)

(reserved)									PMU_HP_SLEEP_DIG_CPU_STALL																	PMU_HP_SLEEP_DIG_PAUSE_WDT																	PMU_HP_SLEEP_DIG_PAD_SLP_SEL																	PMU_HP_SLEEP_HP_PAD_HOLD_ALL																	PMU_HP_SLEEP_LP_PAD_HOLD_ALL																	PMU_HP_SLEEP_UART_WAKEUP_EN																	(reserved)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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PMU_HP_SLEEP_UART_WAKEUP_EN Configures whether to enable UART wake up function in HP_SLEEP state.

0: Disable wake-up function

1: Enable wake-up function

(R/W)

PMU_HP_SLEEP_LP_PAD_HOLD_ALL Configures whether to hold LP GPIO's configuration in HP_SLEEP state.

0: Do not hold

1: Hold

(R/W)

PMU_HP_SLEEP_HP_PAD_HOLD_ALL Configures whether to hold GPIO's configuration in HP_SLEEP state.

0: Do not hold

1: Hold

(R/W)

PMU_HP_SLEEP_DIG_PAD_SLP_SEL Configures whether to use Light-sleep mode configuration for GPIO in HP_SLEEP state.

0: Use normal configuration

1: Use Light-sleep mode configuration. For details please refer to Chapter 9 [GPIO Matrix and IO MUX](#) > Section 9.8 [Pin Functions in Light-sleep](#).

(R/W)

PMU_HP_SLEEP_DIG_PAUSE_WDT Configures whether to pause watchdog in HP_SLEEP state.

0: Do not pause

1: Pause

(R/W)

PMU_HP_SLEEP_DIG_CPU_STALL Configures whether to stall the CPU in HP_SLEEP state.

0: Do not stall

1: Stall

(R/W)

Register 14.14. PMU_HP_SLEEP_HP_CK_POWER_REG (0x007C)

[illegible]

PMU_HP_SLEEP_I2C_ISO_EN Configure whether to power up ANALOG_I2C in HP_SLEEP state.

0: Power down

1: Power up

(R/W)

PMU_HP_SLEEP_I2C_RETENTION Configures whether to enable ANALOG_I2C retention status in HP SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_XPD_PLL_I2C Configures whether to enable PLL I2C controllers in HP_SLEEP state. Each bit controls a PLL I2C register group.

0: Disable

1: Enable

(R/W)

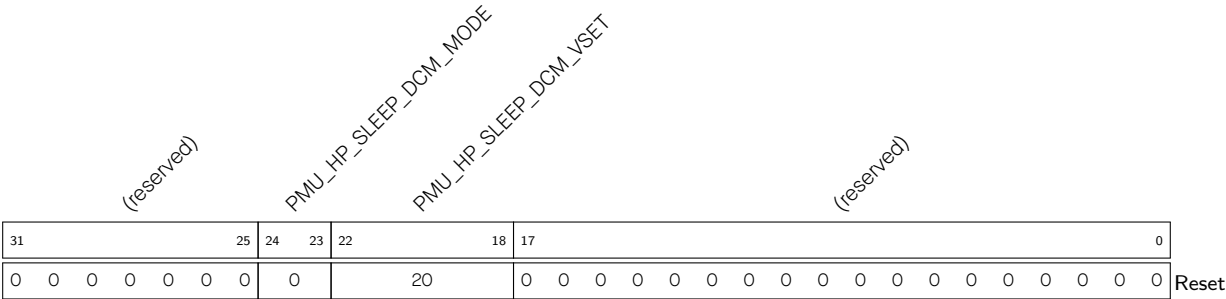
PMU_HP_SLEEP_XPD_PLL Configures whether to enable the analog power domains in HP system in HP_SLEEP state. Bit 0 controls CPLL_CLK, bit 1 controls SPPLL_CLK, bit 2 controls Audio PLL, bit 3 controls SDIO PLL.

0: Disable

1: Enable

(R/W)

Register 14.15. PMU_HP_SLEEP_BIAS_REG (0x0080)



- PMU_HP_SLEEP_DCM_VSET

Regulates the DCDC voltage in HP_SLEEP state. (R/W)
- PMU_HP_SLEEP_DCM_MODE

Configures whether to enable DCDC status in HP_SLEEP state.

0: Disable

1: Enable

2, 3: Reserved

(R/W)

Register 14.16. PMU_HP_SLEEP_BACKUP_REG (0x0084)

[illegible]

PMU_HP_ACTIVE2SLEEP_BACKUP_CLK_SEL Configures the backup module function clock source when PMU state switches from HP ACTIVE to HP SLEEP.

0: Select XTAL

1: Select PLL CLK

2: Select RC_FAST_CLK

3: Invalid value

(R/W)

PMU_HP_ACTIVE2SLEEP_BACKUP_MODE Configures the backup direction and link list when PMU state switches from HP_ACTIVE to HP_SLEEP.

Highest bit:

0: From memory to peripheral

1: From peripheral to memory

Lower two bits:

0: PAU_LINK_ADDR_0

1: PAU_LINK_ADDR_1

2: PAU_LINK_ADDR_2

3: PAU_LINK_ADDR_3

(R/W)

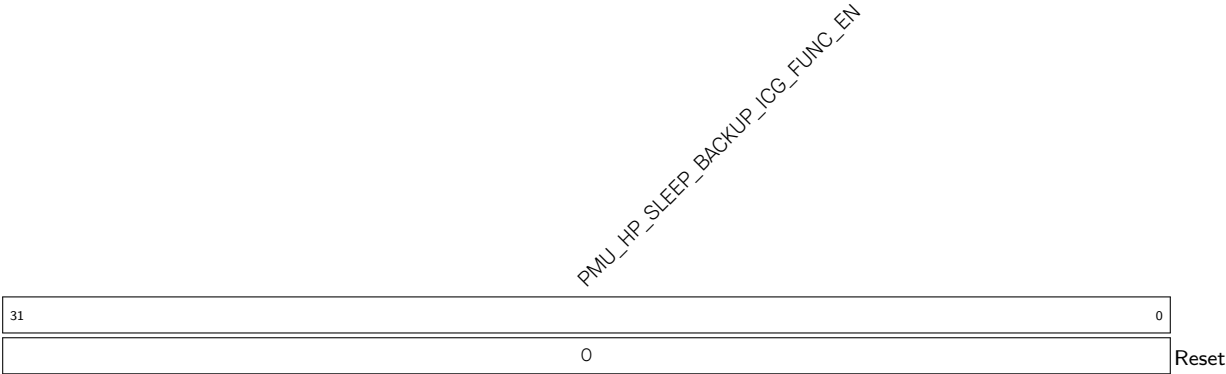
PMU_HP_ACTIVE2SLEEP_BACKUP_EN Configures whether to enable the backup flow when PMU state switches from HP_ACTIVE to HP_SLEEP.

0: Disable backup

1: Enable backup

(R/W)

Register 14.17. PMU_HP_SLEEP_BACKUP_CLK_REG (0x0088)



PMU_HP_SLEEP_BACKUP_ICG_FUNC_EN Configures whether to enable each peripheral's function clock when the target state is HP_SLEEP. For details, please refer to Table 14.4-2.

0: Disable

1: Enable

(R/W)

Register 14.18. PMU_HP_SLEEP_SYSCLK_REG (0x008C)

[illegible]

PMU_HP_SLEEP_ICG_SYS_CLOCK_EN Configures whether to enable ROOT_CLK in HP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_SYS_CLK_SLP_SEL Configures whether to allow PMU to control the clock source in HP_SLEEP state.

0: Controlled by PCR registers

1: Controlled by PMU

(R/W)

PMU_HP_SLEEP_ICG_SLP_SEL Configures whether to allow PMU to control the clock gating in HP SLEEP state.

0: Controlled by PCR registers

1: Controlled by PMU

(R/W)

PMU_HP_SLEEP_DIG_SYS_CLK_SEL Configures the source of ROOT_CLK in HP_SLEEP state.

0: XTAL

1: PLL_CLK

2: RC_FAST_CLK

3: Invalid value

(R/W)

Register 14.19. PMU_HP_SLEEP_HP_REGULATORO_REG (0x0090)

PMU_HP_SLEEP_HP_REGULATOR_DBIAS				(reserved)				PMU_HP_SLEEP_REGULATOR_VO1_XPD				(reserved)				PMU_HP_SLEEP_HP_REGULATOR_XPD				(reserved)															
31	27	26	23	22	21	19	18	17																								0			
24		0 0 0 0		1	0 0 0		1	0 0																								Reset			

PMU_HP_SLEEP_HP_REGULATOR_XPD Configures whether to enable the HP sys regulator in HP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_REGULATOR_VO1_XPD Configures whether to enable VO1 regulator in HP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_HP_REGULATOR_DBIAS Regulates the voltage of the HP sys regulator in HP_SLEEP state. The higher the value, the higher the voltage. (R/W)

Register 14.20. PMU_HP_SLEEP_XTAL_REG (0x0098)

PMU_HP_SLEEP_XPD_XTAL																																	(reserved)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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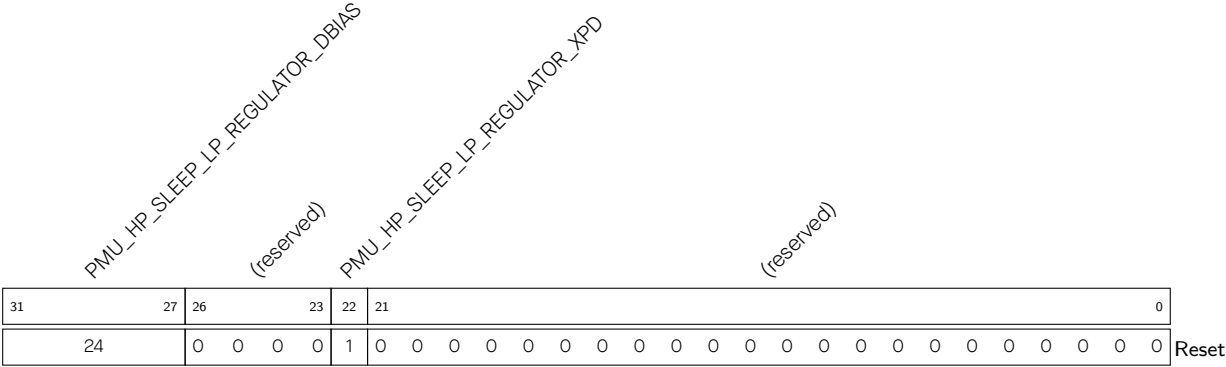
PMU_HP_SLEEP_XPD_XTAL Configures whether to enable XTAL_CLK analog source in HP_SLEEP state.

0: Disable

1: Enable

(R/W)

Register 14.21. PMU_HP_SLEEP_LP_REGULATORO_REG (0x009C)



PMU_HP_SLEEP_LP_REGULATOR_XPD Configures whether to enable LP sys regulator in HP_ACTIVE/HP_SLEEP state.
0: Disable
1: Enable
(R/W)

PMU_HP_SLEEP_LP_REGULATOR_DBIAS Regulates the voltage of the LP sys regulator in HP_ACTIVE/HP_SLEEP state. The higher the value, the higher the voltage. (R/W)

Register 14.22. PMU_HP_SLEEP_LP_DIG_POWER_REG (0x00A8)

PMU_HP_SLEEP_PD_LP_PERI_PD_EN PMU_HP_SLEEP_LP_MEM_DSLP PMU_HP_SLEEP_VDDBAT_MODE PMU_HP_SLEEP_BOD_SOURCE_SEL PMU_HP_SLEEP_LP_PAD_SLP_SEL (reserved)																																	
31	30	29	28	27	26	25																											0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

PMU_HP_SLEEP_LP_PAD_SLP_SEL Configures whether to enable LP GPIO function in HP_ACTIVE/HP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_BOD_SOURCE_SEL Configures whether to enable brown-out detection in HP_ACTIVE/HP_SLEEP. (R/W)

PMU_HP_SLEEP_VDDBAT_MODE Configures whether to enable VBAT power in HP_ACTIVE/HP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_LP_MEM_DSLP Configures whether to enable Memory Deep-sleep mode for the LP SRAM in HP_ACTIVE/HP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_PD_LP_PERI_PD_EN Configures whether to power down LP PD Peripherals in HP_ACTIVE/HP_SLEEP state.

0: Power up

1: Power down

(R/W)

Register 14.23. PMU_HP_SLEEP_LP_CLK_POWER_REG (0x00AC)

PMU_HP_SLEEP_PD_OSC_CLK PMU_HP_SLEEP_XPD_FOSC_CLK PMU_HP_SLEEP_XPD_RC32K PMU_HP_SLEEP_XPD_XTAL32K PMU_HP_SLEEP_XPD_LPPLL (reserved)																																		
31	30	29	28	27	26																													0
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

PMU_HP_SLEEP_XPD_LPPLL Configures whether to enable PLL_LP_CLK in HP_ACTIVE/HP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_XPD_XTAL32K Configures whether to enable XTAL32K_CLK in HP_ACTIVE/HP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_XPD_RC32K Configures whether to enable RC32K_CLK in HP_ACTIVE/HP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_HP_SLEEP_XPD_FOSC_CLK Configures whether to enable RC_FAST_CLK in HP_ACTIVE/HP_SLEEP state.

0: Disable

1: Enable

(R/W)

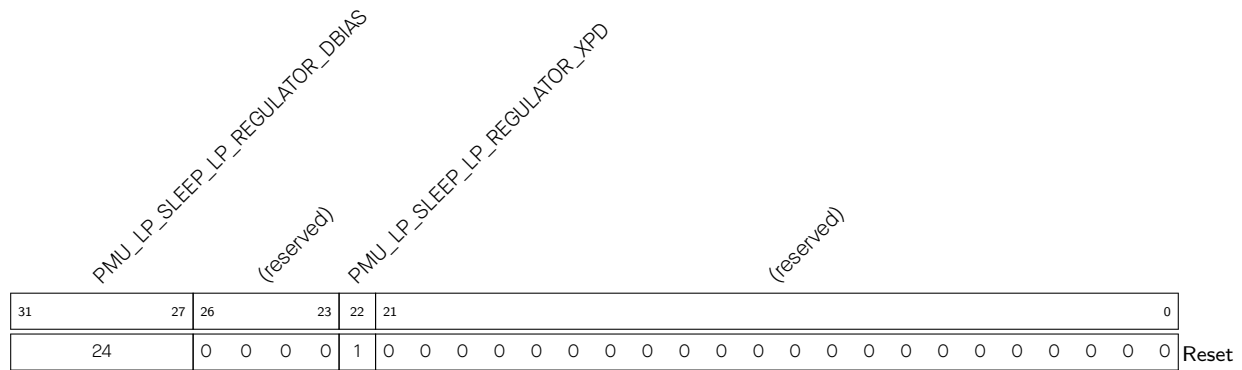
PMU_HP_SLEEP_PD_OSC_CLK Configures whether to enable RC_SLOW_CLK in HP_ACTIVE/HP_SLEEP state.

0: Disable

1: Enable

(R/W)

Register 14.24. PMU_LP_SLEEP_LP_REGULATOR0_REG (0x00B4)



PMU_LP_SLEEP_LP_REGULATOR_XPD Configures whether to enable the LP sys regulator in LP_SLEEP state.

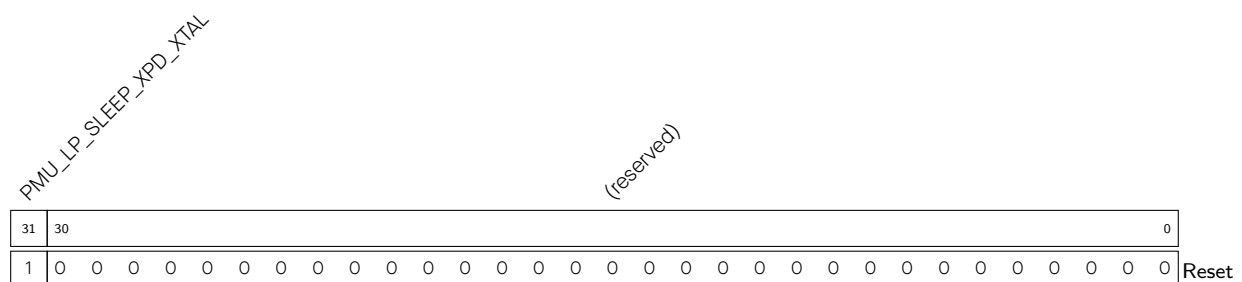
0: Disable

1: Enable

(R/W)

PMU_LP_SLEEP_LP_REGULATOR_DBIAS Regulates the voltage of the LP sys regulator in LP_SLEEP state. The higher the value, the higher the voltage. (R/W)

Register 14.25. PMU_LP_SLEEP_XTAL_REG (0x00BC)



PMU_LP_SLEEP_XPD_XTAL Configures whether to enable XTAL_CLK analog source in LP_SLEEP state.

0: Disable

1: Enable

(R/W)

Register 14.26. PMU_LP_SLEEP_LP_DIG_POWER_REG (0x00C0)

[illegible]

PMU_LP_SLEEP_LP_PAD_SLP_SEL	Configures whether to enable LP GPIO function in LP_SLEEP state. 0: Disable 1: Enable (R/W)
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PMU_LP_SLEEP_BOD_SOURCE_SEL Configures whether to enable brown-out detection in LP_SLEEP state. (R/W)

PMU_LP_SLEEP_VDDBAT_MODE Configures whether to enable VBAT power in LP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_LP_SLEEP_LP_MEM_DSLP	Configures whether to enable Memory Deep-sleep mode for the LP SRAM in LP_SLEEP state.
0: Disable	
1: Enable	
(R/W)	

PMU_LP_SLEEP_PD_LP_PERI_PD_EN Configures whether to power down LP PD Peripherals in LP_SLEEP state.

0: Power up

1: Power down

(R/W)

Register 14.27. PMU_LP_SLEEP_LP_CK_POWER_REG (0x00C4)

[illegible]

PMU_LP_SLEEP_XPD_LPPLL Configures whether to enable PLL_LP_CLK in LP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_LP_SLEEP_XPD_XTAL32K Configures whether to enable XTAL32K_CLK in LP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_LP_SLEEP_XPD_RC32K Configures whether to enable RC32K_CLK in LP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_LP_SLEEP_XPD_FOSC_CLK Configures whether to enable RC_FAST_CLK in LP_SLEEP state.

0: Disable

1: Enable

(R/W)

PMU_LP_SLEEP_PD_OSC_CLK Configures whether to enable RC_SLOW_CLK in LP_SLEEP state.

0: Disable

1: Enable

(R/W)

Register 14.28. PMU_IMM_HP_CK_POWER_REG (0x00CC)

PMU_TIE_HIGH_XPD_XTAL		PMU_TIE_HIGH_XPD_PLL		PMU_TIE_HIGH_XPD_PLL_I2C		PMU_TIE_HIGH_I2C_RETENTION		PMU_TIE_HIGH_GLOBAL_XTAL_ICG		reserved		PMU_TIE_LOW_XPD_XTAL		PMU_TIE_LOW_XPD_PLL		PMU_TIE_LOW_XPD_PLL_I2C		PMU_TIE_LOW_I2C_RETENTION		PMU_TIE_LOW_GLOBAL_XTAL_ICG		reserved	
31	30	27	26	23	22	21	20	17	16	15	14	11	10	7	6	5	4	1	0				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset			

PMU_TIE_LOW_GLOBAL_PLL_ICG Configures whether to power down global PLL clock gating bitmap.

0: No effect

1: Power down

(WT)

PMU_TIE_LOW_GLOBAL_XTAL_ICG Configures whether to power down XTAL clock.

0: No effect

1: Power down

(WT)

PMU_TIE_LOW_I2C_RETENTION Configures whether to power down ANALOG I2C retention.

0: No effect

1: Power down

(WT)

PMU_TIE_LOW_XPD_PLL_I2C Configures whether to power down global PLL I2C clock power bitmap.

0: No effect

1: Power down

(WT)

PMU_TIE_LOW_XPD_PLL Configures whether to power down global PLL clock power bitmap.

0: No effect

1: Power down

(WT)

PMU_TIE_LOW_XPD_XTAL Configures whether to power down XTAL clock power.

0: No effect

1: Power down

(WT)

PMU_TIE_HIGH_GLOBAL_PLL_ICG Configures whether to enable PLL clock gating bitmap.

0: No effect

1: Enable

(WT)

Continued on the next page...

Register 14.28. PMU_IMM_HP_CK_POWER_REG (0x00CC)

Continued from the previous page...

PMU_TIE_HIGH_GLOBAL_XTAL_ICG Configures whether to enable global XTAL clock.

0: No effect

1: Enable

(WT)

PMU_TIE_HIGH_I2C_RETENTION Configures whether to enable ANALOG I2C retention.

0: No effect

1: Enable

(WT)

PMU_TIE_HIGH_XPD_PLL_I2C Configures whether to enable PLL I2C clock power bitmap.

0: No effect

1: Enable

(WT)

PMU_TIE_HIGH_XPD_PLL Configures whether to enable global PLL clock power bitmap.

0: No effect

1: Enable

(WT)

PMU_TIE_HIGH_XPD_XTAL Configures whether to enable global XTAL clock power.

0: No effect

1: Enable

(WT)

Register 14.29. PMU_IMM_SLEEP_SYSCLK_REG (0x00D0)

PMU_UPDATE_DIG_SYS_CLK_SEL				(reserved)																												
31	30	29	28	27																												0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

PMU_UPDATE_DIG_ICG_SWITCH Immediately updates function clock gating status. (WT)

PMU_TIE_LOW_ICG_SLP_SEL Configures whether to disable PMU control of function clock.

0: No effect

1: Disable

(WT)

PMU_TIE_HIGH_ICG_SLP_SEL Configures whether to enable PMU control of function clock.

0: No effect

1: Enable

(WT)

PMU_UPDATE_DIG_SYS_CLK_SEL Immediately updates system clock.

0: No effect

1: Updates the configurations of SYS_CLK_SEL in the current PMU state to the application side.

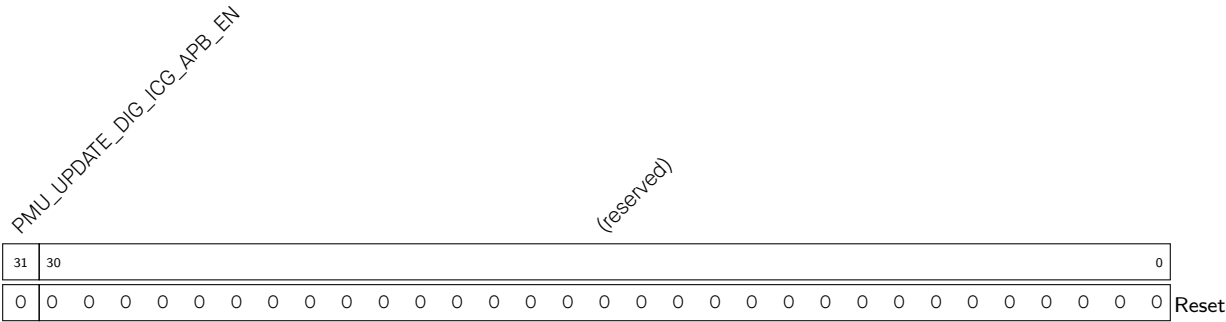
(WT)

Register 14.30. PMU_IMM_HP_FUNC_ICG_REG (0x00D4)

PMU_UPDATE_DIG_ICG_FUNC_EN																																(reserved)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PMU_UPDATE_DIG_ICG_FUNC_EN Immediately updates function clock gating bitmap. (WT)

Register 14.31. PMU_IMM_HP_APB_ICG_REG (0x00D8)



`PMU_UPDATE_DIG_ICG_APB_EN` Immediately updates APB clock gating bitmap. (WT)

Register 14.32. PMU_IMM_PAD_HOLD_ALL_REG (0x00E4)

PMU_TIE_LOW_HP_PAD_HOLD_ALL						(reserved)																					PMU_HP_PAD_HOLD_ALL			
PMU_TIE_HIGH_HP_PAD_HOLD_ALL																										PMU_LP_PAD_HOLD_ALL				
PMU_TIE_LOW_LP_PAD_HOLD_ALL																										PMU_PAD_SLP_SEL				
PMU_TIE_HIGH_LP_PAD_HOLD_ALL																										PMU_PAD_SLP_SEL				
PMU_TIE_LOW_HP_PAD_HOLD_ALL																														
PMU_TIE_HIGH_HP_PAD_HOLD_ALL																														
PMU_TIE_LOW_LP_PAD_HOLD_ALL																														
PMU_TIE_HIGH_LP_PAD_HOLD_ALL																														
31	30	29	28	27	26	25																				3	2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

PMU_PAD_SLP_SEL Indicates whether the current GPIO function uses sleep configurations.

0: No

1: Yes

(RO)

PMU_LP_PAD_HOLD_ALL Indicates whether the LP PAD is in the HOLD status.

0: No

1: Yes

(RO)

PMU_HP_PAD_HOLD_ALL Represents whether the HP PAD is in the HOLD status.

0: No

1: Yes

(RO)

PMU_TIE_HIGH_PAD_SLP_SEL Immediately configures GPIO function to use sleep configurations.

(WT)

PMU_TIE_LOW_PAD_SLP_SEL Immediately releases GPIO function from using sleep configuration.

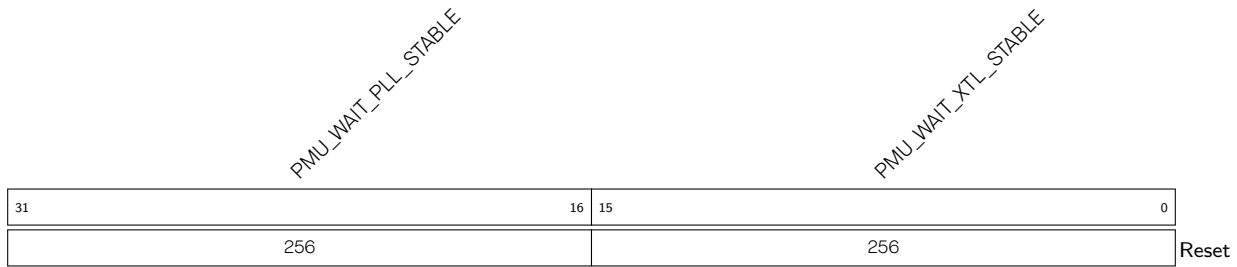
(WT)

PMU_TIE_HIGH_LP_PAD_HOLD_ALL Immediately configure LP PAD to enter HOLD state. (WT)

PMU_TIE_LOW_LP_PAD_HOLD_ALL Immediately releases LP PAD from HOLD state. (WT)

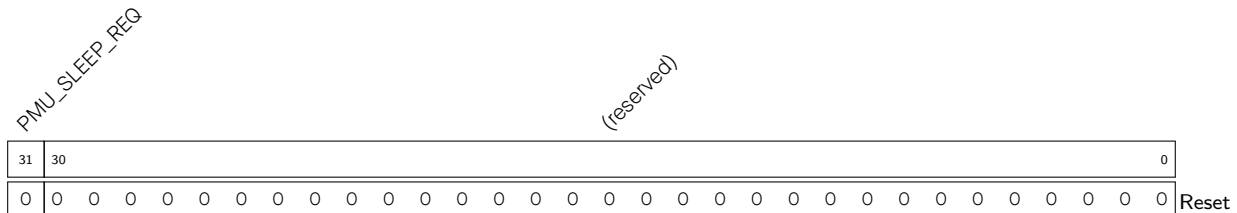
PMU_TIE_HIGH_HP_PAD_HOLD_ALL Immediately configures HP PAD to enter HOLD state. (WT)

PMU_TIE_LOW_HP_PAD_HOLD_ALL Immediately releases HP PAD from HOLD state. (WT)

Register 14.33. PMU_POWER_CK_WAIT_CNTL_REG (0x011C)

PMU_WAIT_XTL_STABLE Configures the wait time for enabling XTAL global clock gating after the XTAL power supply is enabled. The unit is LP_DYN_FAST_CLK. (R/W)

PMU_WAIT_PLL_STABLE Configures the wait time for enabling PLL global clock gating after the PLL power supply is enabled. The unit is LP_DYN_FAST_CLK. (R/W)

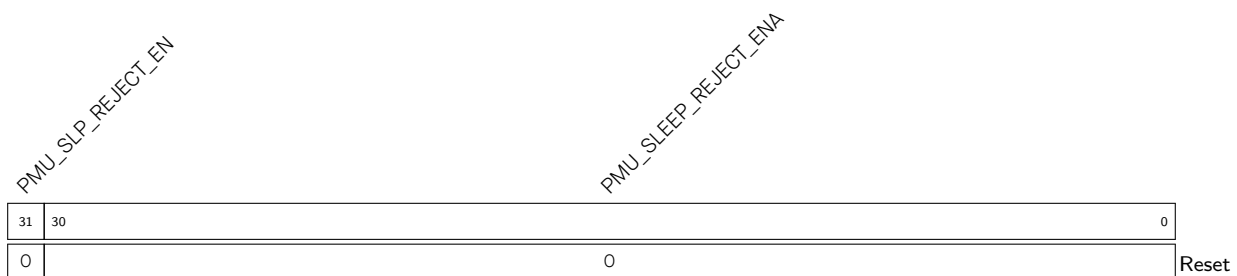
Register 14.34. PMU_SLP_WAKEUP_CNTL0_REG (0x0120)

PMU_SLEEP_REQ Configures whether to switch the PMU state to HP_SLEEP or LP_SLEEP.

0: Do not switch

1: Switch to HP_SLEEP or LP_SLEEP, depending on the state of the LP CPU.

(WT)

Register 14.35. PMU_SLP_WAKEUP_CNTL1_REG (0x0124)

PMU_SLEEP_REJECT_ENA Configures the sleep rejection source. For the mapping between values and sources please refer to Table 14.4-1. (R/W)

PMU_SLP_REJECT_EN Configures whether to enable sleep rejection function.

0: Disable

1: Enable

(R/W)

Register 14.36. PMU_SLP_WAKEUP_CNTL2_REG (0x0128)

31	30	0	
0	0	0	Reset

PMU_WAKEUP_ENA Configures wake-up source. For the mapping between values and sources please refer to Table 14.4-1. (R/W)

Register 14.37. PMU_SLP_WAKEUP_CNTL3_REG (0x012C)

(reserved)																PMU_SLEEP_PRT_SEL		PMU_HP_MIN_SLP_VAL								PMU_LP_MIN_SLP_VAL												
31																18		17	16	15								8		7	0							
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0		0								0								Reset				

PMU_LP_MIN_SLP_VAL Configures the minimum sleep time to enter LP_SLEEP. The unit is LP_DYN_SLOW_CLK. (R/W)

PMU_HP_MIN_SLP_VAL Configures the minimum sleep time to enter HP_SLEEP. The unit is LP DYN SLOW CLK. (R/W)

PMU_SLEEP_PRT_SEL Configures the minimum sleep time mode.

0, 1, 3: Reserved

2: Protects both LP_SLEEP and HP_SLEEP simultaneously

(R/W)

Register 14.38. PMU_SLP_WAKEUP_CNTL4_REG (0x0130)

Diagram illustrating the structure of the `PMU_SLP_REJECT_CAUSE_CLR` register:

- Bit 31: `PMU_SLP_REJECT_CAUSE_CLR`
- Bit 30: `(reserved)`
- Bits 29 down to 0: `Reset`

PMU_SLP_REJECT_CAUSE_CLR Write 1 to clear **PMU_REJECT_CAUSE**. (WT)

Register 14.39. PMU_SLP_WAKEUP_STATUS0_REG (0x0144)

(reserved)																															PMU_WAKEUP_CAUSE																																
31	30																																																														0
0	0																																																														Reset

PMU_WAKEUP_CAUSE Indicates the wake-up source. For the mapping between values and sources please refer to Table 14.4-1. (RO)

Register 14.40. PMU_SLP_WAKEUP_STATUS1_REG (0x0148)

(reserved)																															PMU_REJECT_CAUSE																															
31	30																																																													0
0	0																																																													Reset

PMU_REJECT_CAUSE Indicates the wake-up rejection source. For the mapping between values and sources please refer to Table 14.4-1. (RO)

Register 14.41. PMU_HP_CK_CNTL_REG (0x0154)

(reserved)																PMU_SWITCH_ICG_CNTL_WAIT										PMU_MODIFY_ICG_CNTL_WAIT																																																					
31																15																8																7																0															
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																10																10																Reset																															

PMU_MODIFY_ICG_CNTL_WAIT Configures the wait time for modifying the functional module clock gating. The unit is LP_DYN_FAST_CLK. (R/W)

PMU_SWITCH_ICG_CNTL_WAIT Configures the wait time for switching the functional module clock gating. The unit is LP_DYN_FAST_CLK. (R/W)

Register 14.42. PMU_RF_PWC_REG (0x015C)

(reserved)				PMU_XPD_PERIF_I2C				PMU_PERIF_I2C_RSTB				PMU_SDIO_PLL_XPD				PMU_MSPI_PHY_XPD				(reserved)															
31	28	27	26	25	24	23																													0
0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset		

PMU_MSPI_PHY_XPD Configures whether to enable MSPI PHY.
0: Disable
1: Enable
(R/W)

PMU_SDIO_PLL_XPD Configures whether to enable SDIO PLL.
0: Disable
1: Enable
(R/W)

PMU_PERIF_I2C_RSTB Configures whether to enable reset for PERI I2C.
0: Reset
1: Release
(R/W)

PMU_XPD_PERIF_I2C Configures whether to enable the power for PERI ADC.
0: Disable
1: Enable
(R/W)

Register 14.43. PMU_INT_RAW_REG (0x0164)

[illegible]

PMU_OP1A_CNT_TARGET0_REACH_O_HP_INT_RAW The raw interrupt status of
PMU_OP1A_CNT_TARGET0_REACH_O_INT. (R/WTC/SS)

PMU_OP1A_CNT_TARGET1_REACH_O_HP_INT_RAW The raw interrupt status of
PMU_OP1A_CNT_TARGET1_REACH_O_INT. (R/WTC/SS)

PMU_OP1A_CNT_TARGET0_REACH_1_HP_INT_RAW The raw interrupt status of
PMU_OP1A_CNT_TARGET0_REACH_1_INT. (R/WTC/SS)

PMU_OP1A_CNT_TARGET1_REACH_1_HP_INT_RAW The raw interrupt status of [PMU_OP1A_CNT_TARGET1_REACH_1_INT](#). (R/WTC/SS)

PMU_OP2A_CNT_TARGET0_REACH_O_HP_INT_RAW The raw interrupt status of [PMU_OP2A_CNT_TARGET0_REACH_O_INT](#). (R/WTC/SS)

PMU_OP2A_CNT_TARGET1_REACH_O_HP_INT_RAW The raw interrupt status of [PMU_OP2A_CNT_TARGET1_REACH_O_INT](#). (R/WTC/SS)

PMU_OP2A_CNT_TARGET0_REACH_1_HP_INT_RAW The raw interrupt status of [PMU_OP2A_CNT_TARGET0_REACH_1_INT](#). (R/WTC/SS)

PMU_OP2A_CNT_TARGET1_REACH_1_HP_INT_RAW The raw interrupt status of [PMU_OP2A_CNT_TARGET1_REACH_1_INT](#). (R/WTC/SS)

PMU_LP_CPU_EXC_INT_RAW The raw interrupt status of [PMU_LP_CPU_EXC_INT](#). (R/WTC/SS)

PMU_SDIO_IDLE_INT_RAW The raw interrupt status of [PMU_SDIO_IDLE_INT](#). (R/WTC/SS)

PMU_SW_INT_RAW The raw interrupt status of **PMU_SW_INT**. (R/WTC/SS)

PMU_SOC_SLEEP_REJECT_INT_RAW The raw interrupt status of **PMU_SOC_SLEEP_REJECT_INT**.
(R/WTC/SS)

PMU_SOC_WAKEUP_INT_RAW The raw interrupt status of [PMU_SOC_WAKEUP_INT](#). (R/WTC/SS)

Register 14.44. PMU_HP_INT_ST_REG (0x0168)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PMU_SOC_WAKEUP_INT_ST

PMU_SOC_SLEEP_REJECT_INT_ST

PMU_SW_INT_ST

PMU_SDIO_IDLE_INT_ST

PMU_LP_CPU_EXC_INT_ST

(reserved)

PMU_OP2A_CNT_TARGET1_REACH_1_HP_INT_ST

PMU_OP2A_CNT_TARGET0_REACH_1_HP_INT_ST

PMU_OP2A_CNT_TARGET1_REACH_0_HP_INT_ST

PMU_OP2A_CNT_TARGET0_REACH_0_HP_INT_ST

PMU_OP1A_CNT_TARGET1_REACH_1_HP_INT_ST

PMU_OP1A_CNT_TARGET0_REACH_1_HP_INT_ST

PMU_OP1A_CNT_TARGET1_REACH_0_HP_INT_ST

PMU_OP1A_CNT_TARGET0_REACH_0_HP_INT_ST

(reserved)

Reset

PMU_OP1A_CNT_TARGET0_REACH_0_HP_INT_ST The masked interrupt status of
PMU_OP1A_CNT_TARGET0_REACH_0_INT. (RO)

PMU_OP1A_CNT_TARGET1_REACH_O_HP_INT_ST The masked interrupt status of
PMU_OP1A_CNT_TARGET1_REACH_O_INT. (RO)

PMU_OP1A_CNT_TARGET0_REACH_1_HP_INT_ST The masked interrupt status of
PMU OP1A CNT TARGET0 REACH 1 INT. (RO)

PMU_OP1A_CNT_TARGET1_REACH_1_HP_INT_ST The masked interrupt status of
PMU_OP1A_CNT_TARGET1_REACH_1_INT. (RO)

PMU_OP2A_CNT_TARGETO_REACH_O_HP_INT_ST The masked interrupt status of
PMU OP2A CNT TARGETO REACH O INT. (RO)

PMU_OP2A_CNT_TARGET1_REACH_O_HP_INT_ST The masked interrupt status of
PMU_OP2A_CNT_TARGET1_REACH_O_INT. (RO)

PMU_OP2A_CNT_TARGET0_REACH_1_HP_INT_ST The masked interrupt status of
PMU OP2A CNT TARGET0 REACH 1 INT. (RO)

PMU_OP2A_CNT_TARGET1_REACH_1_HP_INT_ST The masked interrupt status of
PMU OP2A CNT TARGET1 REACH 1 INT. (RO)

PMU LP CPU EXC INT ST The masked interrupt status of PMU LP CPU EXC INT. (RO)

PMU_SDIO_IDLE_INT_ST	The masked interrupt status of PMU_SDIO_IDLE_INT. (RO)
----------------------	--

PMU_SW_INT_ST The masked interrupt status of **PMU_SW_INT**. (RO)

PMU_SOC_SLEEP_REJECT_INT_ST The masked interrupt status of [PMU_SOC_SLEEP_REJECT_INT](#).
(RO)

PMU_SOC_WAKEUP_INT_ST The masked interrupt status of [PMU_SOC_WAKEUP_INT](#). (RO)

Register 14.45. PMU_HP_INT_ENA_REG (0x016C)

PMU_SOC_WAKEUP_INT_ENA PMU_SOC_SLEEP_REJECT_INT_ENA PMU_SW_INT_ENA PMU_SDIO_IDLE_INT_ENA PMU_LP_CPU_EXC_INT_ENA (reserved) reserved reserved reserved reserved PMU_OP2A_CNT_TARGET0_REACH_0_HP_INT_ENA PMU_OP2A_CNT_TARGET0_REACH_1_HP_INT_ENA PMU_OP2A_CNT_TARGET1_REACH_0_HP_INT_ENA PMU_OP2A_CNT_TARGET1_REACH_1_HP_INT_ENA PMU_OP1A_CNT_TARGET0_REACH_0_HP_INT_ENA PMU_OP1A_CNT_TARGET0_REACH_1_HP_INT_ENA PMU_OP1A_CNT_TARGET1_REACH_0_HP_INT_ENA PMU_OP1A_CNT_TARGET1_REACH_1_HP_INT_ENA (reserved)																																			
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13														0			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

PMU_OP1A_CNT_TARGET0_REACH_0_HP_INT_ENA Write 1 to enable
[PMU_OP1A_CNT_TARGET0_REACH_0_INT](#). (R/W)

PMU_OP1A_CNT_TARGET1_REACH_0_HP_INT_ENA Write 1 to enable
[PMU_OP1A_CNT_TARGET1_REACH_0_INT](#). (R/W)

PMU_OP1A_CNT_TARGET0_REACH_1_HP_INT_ENA Write 1 to enable
[PMU_OP1A_CNT_TARGET0_REACH_1_INT](#). (R/W)

PMU_OP1A_CNT_TARGET1_REACH_1_HP_INT_ENA Write 1 to enable
[PMU_OP1A_CNT_TARGET1_REACH_1_INT](#). (R/W)

PMU_OP2A_CNT_TARGET0_REACH_0_HP_INT_ENA Write 1 to enable
[PMU_OP2A_CNT_TARGET0_REACH_0_INT](#). (R/W)

PMU_OP2A_CNT_TARGET1_REACH_0_HP_INT_ENA Write 1 to enable
[PMU_OP2A_CNT_TARGET1_REACH_0_INT](#). (R/W)

PMU_OP2A_CNT_TARGET0_REACH_1_HP_INT_ENA Write 1 to enable
[PMU_OP2A_CNT_TARGET0_REACH_1_INT](#). (R/W)

PMU_OP2A_CNT_TARGET1_REACH_1_HP_INT_ENA Write 1 to enable
[PMU_OP2A_CNT_TARGET1_REACH_1_INT](#). (R/W)

PMU_LP_CPU_EXC_INT_ENA Write 1 to enable [PMU_LP_CPU_EXC_INT](#). (R/W)

PMU_SDIO_IDLE_INT_ENA Write 1 to enable [PMU_SDIO_IDLE_INT](#). (R/W)

PMU_SW_INT_ENA Write 1 to enable [PMU_SW_INT](#). (R/W)

PMU_SOC_SLEEP_REJECT_INT_ENA Write 1 to enable [PMU_SOC_SLEEP_REJECT_INT](#). (R/W)

PMU_SOC_WAKEUP_INT_ENA Write 1 to enable [PMU_SOC_WAKEUP_INT](#). (R/W)

Register 14.46. PMU_HP_INT_CLR_REG (0x0170)

[illegible]

PMU_OP1A_CNT_TARGETO_REACH_O_HP_INT_CLR Write 1 to clear
PMU_OP1A_CNT_TARGETO_REACH_O_INT. (WT)

PMU_OP1A_CNT_TARGET1_REACH_O_HP_INT_CLR Write 1 to clear
PMU_OP1A_CNT_TARGET1_REACH_O_INT. (WT)

PMU_OP1A_CNT_TARGET0_REACH_1_HP_INT_CLR Write 1 to clear
PMU_OP1A_CNT_TARGET0_REACH_1_INT. (WT)

PMU_OP1A_CNT_TARGET1_REACH_1_HP_INT_CLR Write 1 to clear
PMU_OP1A_CNT_TARGET1_REACH_1_INT. (WT)

PMU_OP2A_CNT_TARGETO_REACH_O_HP_INT_CLR Write 1 to clear
VPMU_OP2A_CNT_TARGETO_REACH_O_INT. (WT)

PMU_OP2A_CNT_TARGET1_REACH_O_HP_INT_CLR Write 1 to clear
PMU_OP2A_CNT_TARGET1_REACH_O_INT. (WT)

PMU_OP2A_CNT_TARGET0_REACH_1_HP_INT_CLR Write 1 to clear
PMU_OP2A_CNT_TARGET0_REACH_1_INT. (WT)

PMU_OP2A_CNT_TARGET1_REACH_1_HP_INT_CLR Write 1 to clear
PMU_OP2A_CNT_TARGET1_REACH_1_INT. (WT)

PMU_LP_CPU_EXC_INT_CLR Write 1 to clear **PMU_LP_CPU_EXC_INT**. (WT)

PMU_SDIO_IDLE_INT_CLR Write 1 to clear [PMU_SDIO_IDLE_INT](#). (WT)

PMU_SW_INT_CLR Write 1 to clear **PMU_SW_INT**. (WT)

PMU_SOC_SLEEP_REJECT_INT_CLR Write 1 to clear **PMU_SOC_SLEEP_REJECT_INT**. (WT)

PMU SOC WAKEUP INT CLR Write 1 to clear **PMU SOC WAKEUP INT.** (WT)

Register 14.47. PMU_LP_INT_RAW_REG (0x0174)

[illegible]

PMU_LP_CPU_SLEEP_REJECT_INT_RAW The raw interrupt status of [PMU_LP_CPU_SLEEP_REJECT_INT](#). (R/WTC/SS)

PMU_OP1A_CNT_TARGETO_REACH_O_LP_INT_RAW The raw interrupt status of [PMU_OP1A_CNT_TARGETO_REACH_O_LP_INT](#). (R/WTC/SS)

PMU_OP1A_CNT_TARGET1_REACH_O_LP_INT_RAW The raw interrupt status of [PMU_OP1A_CNT_TARGET1_REACH_O_LP_INT](#). (R/WTC/SS)

PMU_OP1A_CNT_TARGET0_REACH_1_LP_INT_RAW The raw interrupt status of
PMU_OP1A_CNT_TARGET0_REACH_1_LP_INT. (R/WTC/SS)

PMU_OP1A_CNT_TARGET1_REACH_1_LP_INT_RAW The raw interrupt status of [PMU_OP1A_CNT_TARGET1_REACH_1_LP_INT](#). (R/WTC/SS)

PMU_OP2A_CNT_TARGETO_REACH_O_LP_INT_RAW The raw interrupt status of [PMU_OP2A_CNT_TARGETO_REACH_O_LP_INT](#). (R/WTC/SS)

PMU_OP2A_CNT_TARGET1_REACH_O_LP_INT_RAW The raw interrupt status of [PMU_OP2A_CNT_TARGET1_REACH_O_LP_INT](#). (R/WTC/SS)

PMU_OP2A_CNT_TARGETO_REACH_1_LP_INT_RAW The raw interrupt status of [PMU_OP2A_CNT_TARGETO_REACH_1_LP_INT](#). (R/WTC/SS)

PMU_OP2A_CNT_TARGET1_REACH_1_LP_INT_RAW The raw interrupt status of
PMU_OP2A_CNT_TARGET1_REACH_1_LP_INT. (R/WTC/SS)

PMU_LP_CPU_WAKEUP_INT_RAW The raw interrupt status of [PMU_LP_CPU_WAKEUP_INT](#).
(R/WTC/SS)

PMU_SLEEP_SWITCH_ACTIVE_END_INT_RAW The raw interrupt status of
PMU_SLEEP_SWITCH_ACTIVE_END_INT. (R/WTC/SS)

PMU_ACTIVE_SWITCH_SLEEP_END_INT_RAW	The raw interrupt status of PMU ACTIVE SWITCH SLEEP END INT. (R/WTC/SS)
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PMU_SLEEP_SWITCH_ACTIVE_START_INT_RAW	The raw interrupt status of
PMU_SLEEP_SWITCH_ACTIVE_START_INT. (R/WTC/SS)	

Continued on the next page...

Register 14.47. PMU_LP_INT_RAW_REG (0x0174)

Continued from the previous page...

PMU_ACTIVE_SWITCH_SLEEP_START_INT_RAW The raw interrupt status of [PMU_ACTIVE_SWITCH_SLEEP_START_INT](#). (R/WTC/SS)

PMU_HP_SW_TRIGGER_INT_RAW The raw interrupt status of [PMU_HP_SW_TRIGGER_INT](#). (R/WTC/SS)

Register 14.48. PMU_LP_INT_ST_REG (0x0178)

[illegible]

PMU_LP_CPU_SLEEP_REJECT_INT_ST The masked interrupt status of [PMU_LP_CPU_SLEEP_REJECT_INT](#). (RO)

PMU_OP1A_CNT_TARGETO_REACH_O_LP_INT_ST The masked interrupt status of [PMU_OP1A_CNT_TARGETO_REACH_O_LP_INT](#). (RO)

PMU_OP1A_CNT_TARGET1_REACH_O_LP_INT_ST The masked interrupt status of [PMU_OP1A_CNT_TARGET1_REACH_O_LP_INT](#). (RO)

PMU_OP1A_CNT_TARGET0_REACH_1_LP_INT_ST The masked interrupt status of [PMU_OP1A_CNT_TARGET0_REACH_1_LP_INT](#). (RO)

PMU_OP1A_CNT_TARGET1_REACH_1_LP_INT_ST	The masked interrupt status of PMU_OP1A_CNT_TARGET1_REACH_1_LP_INT. (RO)
--	--

PMU_OP2A_CNT_TARGETO_REACH_O_LP_INT_ST	The masked interrupt status of PMU_OP2A_CNT_TARGETO_REACH_O_LP_INT. (RO)
--	--

PMU_OP2A_CNT_TARGET1_REACH_O_LP_INT_ST The masked interrupt status of [PMU_OP2A_CNT_TARGET1_REACH_O_LP_INT](#). (RO)

PMU_OP2A_CNT_TARGET0_REACH_1_LP_INT_ST The masked interrupt status of [PMU_OP2A_CNT_TARGET0_REACH_1_LP_INT](#). (RO)

PMU_OP2A_CNT_TARGET1_REACH_1_LP_INT_ST	The masked interrupt status of PMU_OP2A_CNT_TARGET1_REACH_1_LP_INT. (RO)
--	--

PMU_LP_CPU_WAKEUP_INT_ST The masked interrupt status of [PMU_LP_CPU_WAKEUP_INT](#). (RO)

PMU_SLEEP_SWITCH_ACTIVE_END_INT_ST The masked interrupt status of [PMU_SLEEP_SWITCH_ACTIVE_END_INT](#). (RO)

PMU_ACTIVE_SWITCH_SLEEP_END_INT_ST The masked interrupt status of PMU ACTIVE SWITCH SLEEP END INT. (RO)

PMU_SLEEP_SWITCH_ACTIVE_START_INT_ST	The masked interrupt status of PMU SLEEP SWITCH ACTIVE START INT. (RO)
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Continued on the next page...

Register 14.48. PMU_LP_INT_ST_REG (0x0178)

Continued from the previous page...

PMU_ACTIVE_SWITCH_SLEEP_START_INT_ST The masked interrupt status of [PMU_ACTIVE_SWITCH_SLEEP_START_INT](#). (RO)

PMU_HP_SW_TRIGGER_INT_ST The masked interrupt status of [PMU_HP_SW_TRIGGER_INT](#). (RO)

Register 14.49. PMU_LP_INT_ENA_REG (0x017C)

[illegible]

PMU_LP_CPU_SLEEP_REJECT_INT_ENA Write 1 to enable [PMU_LP_CPU_SLEEP_REJECT_INT](#).
(R/W)

PMU_OP1A_CNT_TARGET0_REACH_O_LP_INT_ENA	Write	1	to	enable
PMU_OP1A_CNT_TARGET0_REACH_O_LP_INT. (R/W)				

PMU_OP1A_CNT_TARGET1_REACH_O_LP_INT_ENA	Write	1	to	enable
PMU_OP1A_CNT_TARGET1_REACH_O_LP_INT. (R/W)				

PMU_OP1A_CNT_TARGET0_REACH_1_LP_INT_ENA	Write	1	to	enable
PMU OP1A CNT TARGET0 REACH 1 LP INT. (R/W)				

PMU_OP1A_CNT_TARGET1_REACH_1_LP_INT_ENA	Write	1	to	enable
PMU_OP1A_CNT_TARGET1_REACH_1_LP_INT. (R/W)				

PMU_OP2A_CNT_TARGETO_REACH_O_LP_INT_ENA	Write	1	to	enable
PMU_OP2A_CNT_TARGETO_REACH_O_LP_INT. (R/W)				

PMU_OP2A_CNT_TARGET1_REACH_O_LP_INT_ENA	Write	1	to	enable
PMU OP2A CNT TARGET1 REACH O LP INT. (R/W)				

PMU_OP2A_CNT_TARGET0_REACH_1_LP_INT_ENA	Write	1	to	enable
PMU OP2A CNT TARGET0 REACH 1 LP INT. (R/W)				

PMU_OP2A_CNT_TARGET1_REACH_1_LP_INT_ENA	Write	1	to	enable
PMU_OP2A_CNT_TARGET1_REACH_1_LP_INT. (R/W)				

PMU LP CPU WAKEUP INT ENA Write 1 to enable **PMU LP CPU WAKEUP INT**. (R/W)

PMU_SLEEP_SWITCH_ACTIVE_END_INT_ENA Write 1 to enable [PMU_SLEEP_SWITCH_ACTIVE_END_INT](#).
(R/W)

PMU_ACTIVE_SWITCH_SLEEP_END_INT_ENA Write 1 to enable [PMU_ACTIVE_SWITCH_SLEEP_END_INT](#).
(R/W)

PMU_SLEEP_SWITCH_ACTIVE_START_INT_ENA	Write	1	to	enable
PMU_SLEEP_SWITCH_ACTIVE_START_INT. (R/W)				

Continued on the next page...

Register 14.49. PMU_LP_INT_ENA_REG (0x017C)

Continued from the previous page...

PMU_ACTIVE_SWITCH_SLEEP_START_INT_ENA Write 1 to enable
[PMU_ACTIVE_SWITCH_SLEEP_START_INT](#). (R/W)

PMU_HP_SW_TRIGGER_INT_ENA Write 1 to enable [PMU_HP_SW_TRIGGER_INT](#). (R/W)

Register 14.50. PMU_LP_INT_CLR_REG (0x0180)

[illegible]

PMU_LP_CPU_SLEEP_REJECT_LP_INT_CLR Write 1 to clear **PMU_LP_CPU_SLEEP_REJECT_INT**.
(R/W)

PMU_OP1A_CNT_TARGET0_REACH_O_LP_INT_CLR Write 1 to clear
PMU_OP1A_CNT_TARGET0_REACH_O_LP_INT. (WT)

PMU_OP1A_CNT_TARGET1_REACH_O_LP_INT_CLR Write 1 to clear
PMU_OP1A_CNT_TARGET1_REACH_O_LP_INT. (WT)

PMU_OP1A_CNT_TARGET0_REACH_1_LP_INT_CLR Write 1 to clear
PMU_OP1A_CNT_TARGET0_REACH_1_LP_INT. (WT)

PMU_OP1A_CNT_TARGET1_REACH_1_LP_INT_CLR Write 1 to clear
PMU_OP1A_CNT_TARGET1_REACH_1_LP_INT. (WT)

PMU_OP2A_CNT_TARGET0_REACH_O_LP_INT_CLR Write 1 to clear
PMU_OP2A_CNT_TARGET0_REACH_O_LP_INT. (WT)

PMU_OP2A_CNT_TARGET1_REACH_O_LP_INT_CLR Write 1 to clear
PMU_OP2A_CNT_TARGET1_REACH_O_LP_INT. (WT)

PMU_OP2A_CNT_TARGET0_REACH_1_LP_INT_CLR Write 1 to clear
PMU_OP2A_CNT_TARGET0_REACH_1_LP_INT. (WT)

PMU_OP2A_CNT_TARGET1_REACH_1_LP_INT_CLR Write 1 to clear
PMU_OP2A_CNT_TARGET1_REACH_1_LP_INT. (WT)

PMU_LP_CPU_WAKEUP_INT_CLR Write 1 to clear **PMU_LP_CPU_WAKEUP_INT**. (WT)

PMU_SLEEP_SWITCH_ACTIVE_END_INT_CLR Write 1 to clear [PMU_SLEEP_SWITCH_ACTIVE_END_INT](#).
(WT)

PMU_ACTIVE_SWITCH_SLEEP_END_INT_CLR Write 1 to clear [PMU_ACTIVE_SWITCH_SLEEP_END_INT](#).
(WT)

PMU_SLEEP_SWITCH_ACTIVE_START_INT_CLR Write 1 to clear **PMU_SLEEP_SWITCH_ACTIVE_START_INT**.
(WT)

Continued on the next page...

Register 14.50. PMU_LP_INT_CLR_REG (0x0180)

Continued from the previous page...

PMU_ACTIVE_SWITCH_SLEEP_START_INT_CLR Write 1 to clear [PMU_ACTIVE_SWITCH_SLEEP_START_INT](#).
(WT)

PMU_HP_SW_TRIGGER_INT_CLR Write 1 to clear [PMU_HP_SW_TRIGGER_INT](#). (WT)

Register 14.51. PMU_LP_CPU_PWRO_REG (0x0184)

PMU_LP_CPU_SLP_BYPASS_INTR_EN				PMU_LP_CPU_SLP_STALL_WAIT								(reserved)								PMU_LP_CPU_STALL_RDY																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
PMU_LP_CPU_SLP_RESET_EN				PMU_LP_CPU_SLP_WAITI_FLAG_EN								PMU_LP_CPU_FORCE_STALL								PMU_LP_CPU_WAITI_RDY																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
PMU_LP_CPU_SLP_STALL_EN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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PMU_LP_CPU_WAITI_RDY Indicates whether the LP CPU is in the waiti state.

0: Does not enter the state

1: Enters the state

(RO)

PMU_LP_CPU_STALL_RDY Indicates whether the LP CPU is in the stall state.

0: Does not enter the state

1: Enters the state

(RO)

PMU_LP_CPU_FORCE_STALL Configures the LP CPU to enter the stall state.

0: Does not enter the state

1: Enters the state

(R/W)

PMU_LP_CPU_SLP_WAITI_FLAG_EN Configures whether the LP CPU needs to enter the waiti state when entering sleep mode.

0: Does not enter the state

1: Enters the state

(R/W)

PMU_LP_CPU_SLP_STALL_FLAG_EN Configures whether the LP CPU needs to enter the stall state when entering sleep mode.

0: Does not enter the state

1: Enters the state

(R/W)

PMU_LP_CPU_SLP_STALL_WAIT Configures the time to wait for the stall to take effect after enabling stall state when the LP CPU enters sleep mode. The unit is LP_DYN_FAST_CLK. (R/W)

PMU_LP_CPU_SLP_STALL_EN Configures whether to pause the LP CPU in sleep mode.

0: Do not pause

1: Pause

(R/W)

Continued on the next page...

Register 14.51. PMU_LP_CPU_PWRO_REG (0x0184)

Continued from the previous page...

PMU_LP_CPU_SLP_RESET_EN Configures whether to reset the LP CPU in sleep mode.

0: Do not reset

1: Reset

(R/W)

PMU_LP_CPU_SLP_BYPASS_INTR_EN Configures whether to enable interrupt signals for the LP CPU in sleep mode.

0: Disable interrupt signals

1: Enable interrupt signals

(R/W)

Register 14.52. PMU_LP_CPU_PWR1_REG (0x0188)

[illegible]

PMU_LP_CPU_SLEEP_REQ Configures whether the LP CPU enters sleep.

0: Does not enter sleep

1: Enters sleep

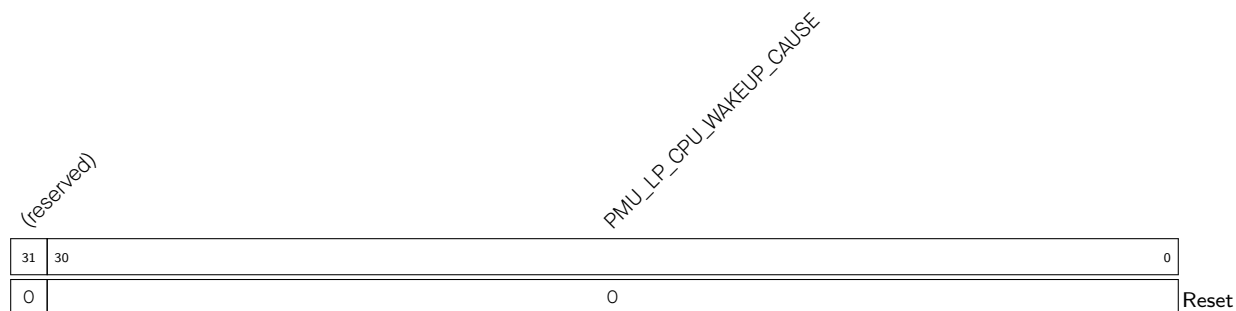
(WT)

Register 14.53. PMU_LP_CPU_PWR2_REG (0x018C)

Diagram illustrating the structure of the `PMU_LP_CPU_WAKEUP_EN` register. The register is 32 bits wide. Bit 31 is labeled `(reserved)`. Bit 30 is labeled `PMU_LP_CPU_WAKEUP_EN`. Bit 0 is labeled `Reset`.

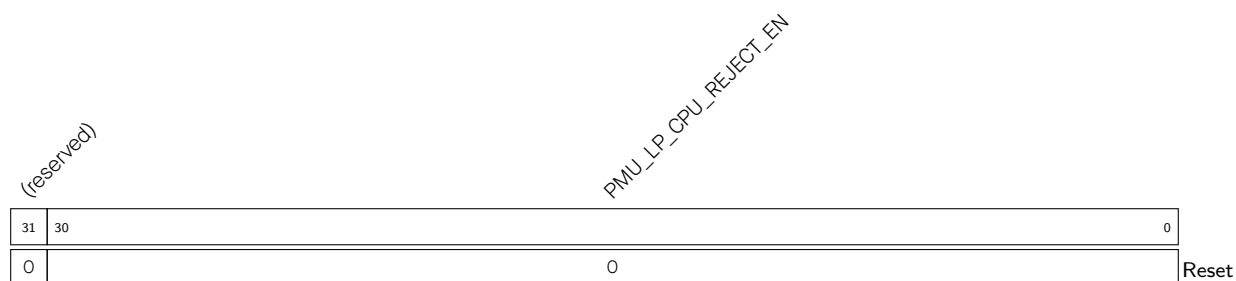
PMU_LP_CPU_WAKEUP_EN Configures the wake-up source for LP CPU. For details please refer to Table 14.4-1 *Wake-up Sources*. (R/W)

Register 14.54. PMU_LP_CPU_PWR3_REG (0x0190)



PMU_LP_CPU_WAKEUP_CAUSE Indicates the wake-up cause of LP CPU. (RO)

Register 14.55. PMU_LP_CPU_PWR4_REG (0x0194)



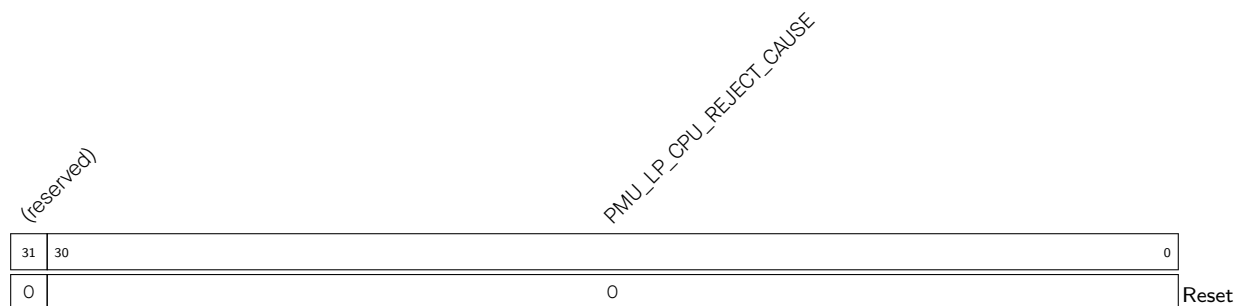
PMU_LP_CPU_REJECT_EN Configures whether to enable sleep rejection function for the LP CPU.

0: Disable

1: Enable

(R/W)

Register 14.56. PMU_LP_CPU_PWR5_REG (0x0198)



PMU_LP_CPU_REJECT_CAUSE Indicates the sleep rejection cause of LP CPU. (RO)

PMU_HP_TRIGGER_LP
PMU_LP_TRIGGER_HP

[illegible]

PMU_HP_TRIGGER_LP When the HP CPU sets this register to 1, the LP CPU is woken up. (WT)

reserved

PMU_OP1A_TARGET0_0

PMU_OP1A_TARGET1_0

PMU_OP1A_TIEH_0

reserved

PMU_OP1A_TIEH_SEL_0

reserved

PMU_OP1A_FORCE_TIEH_SEL_0

(reserved)

31	30	23	22	15	14	13	12	11	9	8	7	6	0					
0	0x80		0x40			0	0	0	0	1	0	0	0	0	0	0	0	Reset

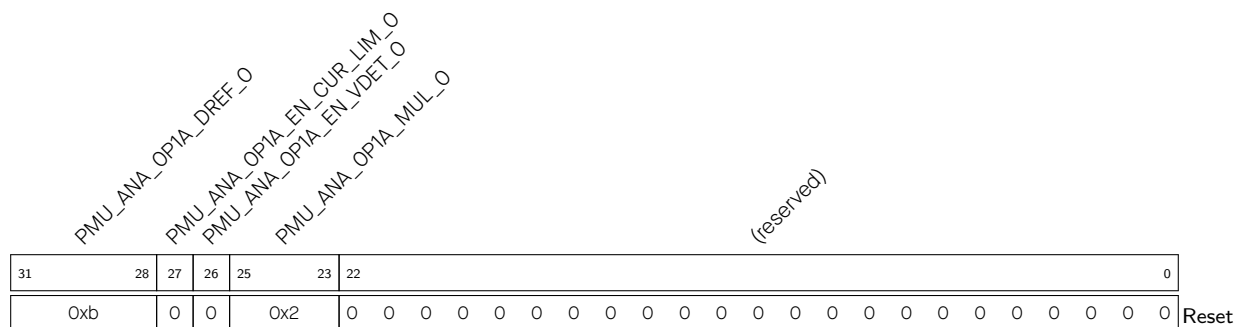
(R/W)

(R/W)

(R/W)

PMU_OP1A_TARGETO_0 Configures the timeout for stage 1 of the VO1 regulator wait counter, with the unit being a 16 division of LP_DYN_SLOW_CLK. (R/W)

Register 14.59. PMU_EXT_LDO_PO_OP1A_ANA_REG (0x01BC)



PMU_ANA_OP1A_MUL_0 Configures the MUL, where the output voltage of the regulator is $MUL \times V_{ref}$.

0-7: MUL ranges from 1 to 2.75, with a step of 0.25.
(R/W)

PMU_ANA_OP1A_EN_VDET_0 Configures whether to enable V01 voltage detection.

0: Do not enable
1: Enable
(R/W)

PMU_ANA_OP1A_EN_CUR_LIM_0 Configures whether to enable over current protection for VO1.

0: Do not enable
1: Enable
(R/W)

PMU_ANA_OP1A_DREF_0 Configures the LDO Vref, where the output voltage of the regulator is $MUL \times V_{ref}$.

0-8: $V_{ref} = 0.5 \text{ V} - 0.9 \text{ V}$, with a step of 50 mV
9-15: $V_{ref} = 1.0 \text{ V} - 1.6 \text{ V}$, with a step of 100 mV
(R/W)

Register 14.60. PMU_EXT_LDO_PO_OP2A_REG (0x01C0)

[illegible]

PMU_OP2A_FORCE_TIEH_SEL_0 Configures which of the following controls the mode selection for VO3 regulator.

0: Mode selection is controlled by `PMU_OP2A_TIEH_SEL_0`

1: Mode selection is controlled by SDMMC 0

2: Bypass mode

3: Mode selection is controlled by SDMMC_1

(R/W)

PMU_OP2A_XPD_0 Configures whether software enables V03 regulator.

0: Disable

1: Enable

(R/W)

PMU_OP2A_TIEH_SEL_0 Configures mode for VO3.

0: Mode is controlled by PMU_OP2A_TIEH_0.

1: Mode is controlled by SDMMC_0

2: Bypass mode

3: Mode is controlled by SDMMC_1

(R/W)

PMU_OP2A_TIEH_0 Configures mode for VO3.

0: LDO mode

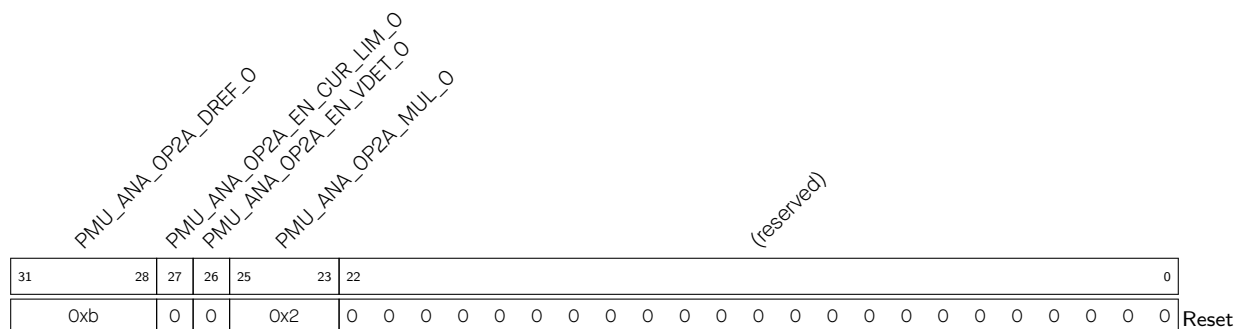
1: Bypass mode

(R/W)

PMU_OP2A_TARGET1_0 Configures the timeout for stage 2 of the VO3 regulator wait counter, with the unit being a 16 division of LP_DYN_SLOW_CLK. (R/W)

PMU_OP2A_TARGET0_0 Configures the timeout for stage 1 of the VO3 regulator wait counter, with the unit being a 16 division of LP_DYN_SLOW_CLK. (R/W)

Register 14.61. PMU_EXT_LDO_PO_OP2A_ANA_REG (0x01C4)



PMU_ANA_OP2A_MUL_0 Configures the MUL, where the output voltage of the regulator is $MUL \times V_{ref}$.

0-7: MUL ranges from 1 to 2.75, with a step of 0.25.

(R/W)

PMU_ANA_OP2A_EN_VDET_0 Configures whether to enable VO3 voltage detection.

0: Do not enable

1: Enable

(R/W)

PMU_ANA_OP2A_EN_CUR_LIM_0 Configures whether to enable over current protection for V03.

0: Do not enable

1: Enable

(R/W)

PMU_ANA_OP2A_DREF_0 Configures the LDO Vref, where the output voltage of the regulator is $MUL \times V_{ref}$.

0-8: $V_{ref} = 0.5 \text{ V} - 0.9 \text{ V}$, with a step of 50 mV

9-15: $V_{ref} = 1.0\text{ V} - 1.6\text{ V}$, with a step of 100 mV

(R/W)

Register 14.62. PMU_EXT_LDO_P1_OP1A_REG (0x01D0)

[illegible]

PMU_OP1A_FORCE_TIEH_SEL_1 Configures which of the following controls the mode selection for VO2 regulator.

0: Mode selection is controlled by `PMU_OP1A_TIEH_SEL_1`

1: Mode selection is controlled by SDMMC_0

2: Bypass mode

3: Mode selection is controlled by SDMMC_1

(R/W)

PMU_OP1A_XPD_1 Configures whether software enables VO2 regulator.

0: Disable

1: Enable

(R/W)

PMU_OP1A_TIEH_SEL_1 Configures mode for VO2.

0: Mode is controlled by PMU_OP1A_TIEH_1.

1: Mode is controlled by SDMMC_0

2: Bypass mode

3: Mode is controlled by SDMMC_1

(R/W)

PMU_OP1A_TIEH_1 Configures mode for V02.

0: LDO mode

1: Bypass mode

(R/W)

PMU_OP1A_TARGET1_1 Configures the timeout for stage 2 of the VO2 regulator wait counter, with the unit being a 16 division of LP_DYN_SLOW_CLK. (R/W)

PMU_OP1A_TARGET0_1 Configures the timeout for stage 1 of the VO2 regulator wait counter, with the unit being a 16 division of LP_DYN_SLOW_CLK. (R/W)

Register 14.63. PMU_EXT_LDO_P1_OP1A_ANA_REG (0x01D4)

Diagram illustrating the structure of the PMU_Ana_Op1A register (32 bits wide). The register is divided into four 8-bit fields:

- PMU_Ana_Op1A_DREF_1 (bits 31-28)
- PMU_Ana_Op1A_EN_CUR_LIM_1 (bits 27-26)
- PMU_Ana_Op1A_EN_VDET_1 (bits 25-23)
- PMU_Ana_Op1A_MUL_1 (bits 22-20)

The remaining bits (19-0) are reserved.

PMU_ANA_OP1A_MUL_1 Configures the MUL, where the output voltage of the regulator is $MUL \times V_{ref}$.

0-7: MUL ranges from 1 to 2.75, with a step of 0.25.

(R/W)

PMU_ANA_OP1A_EN_VDET_1 Configures whether to enable V02 voltage detection.

0: Do not enable

1: Enable

(R/W)

PMU_ANA_OP1A_EN_CUR_LIM_1	Configures whether to enable over current protection for VO2.
---------------------------	---

0: Do not enable

1: Enable

(R/W)

PMU_ANA_OP1A_DREF_1 Configures the LDO Vref, where the output voltage of the regulator is MUL × Vref.

0-8: $V_{ref} = 0.5 \text{ V} - 0.9 \text{ V}$, with a step of 50 mV

9-15: $V_{ref} = 1.0 \text{ V} - 1.6 \text{ V}$, with a step of 100 mV

(R/W)

Register 14.64. PMU_EXT_LDO_P1_OP2A_REG (0x01D8)

reserved		PMU_OP2A_TARGET0_1												PMU_OP2A_TARGET1_1												PMU_OP2A_TIEH_1				reserved		PMU_OP2A_TIEH_SEL_1				PMU_OP2A_XPD_1				PMU_OP2A_FORCE_TIEH_SEL_1				(reserved)			
31	30	23												22	15												14	13	12	11	9		8	7	6	0											
0	0x80												0x40												0	0	0	0		1	0	0	0	0	0	0	0	0	0	0	0	0	Reset				

PMU_OP2A_FORCE_TIEH_SEL_1 Configures which of the following controls the mode selection for VO4 regulator.

- 1: Mode selection is controlled by [PMU_OP2A_TIEH_SEL_1](#)
 - 1: Mode selection is controlled by SDMMC_0
 - 2: Bypass mode
 - 3: Mode selection is controlled by SDMMC_1
- (R/W)

PMU_OP2A_XPD_1 Configures whether software enables VO4 regulator.

- 0: Disable
 - 1: Enable
- (R/W)

PMU_OP2A_TIEH_SEL_1 Configures mode for VO4.

- 0: Mode is controlled by [PMU_OP2A_TIEH_1](#).
 - 1: Mode is controlled by SDMMC_0
 - 2: Bypass mode
 - 3: Mode is controlled by SDMMC_1
- (R/W)

PMU_OP2A_TIEH_1 Configures mode for VO4.

- 0: LDO mode
 - 1: Bypass mode
- (R/W)

PMU_OP2A_TARGET1_1 Configures the timeout for stage 2 of the VO4 regulator wait counter, with the unit being a 16 division of LP_DYN_SLOW_CLK. (R/W)

PMU_OP2A_TARGET0_1 Configures the timeout for stage 1 of the VO4 regulator wait counter, with the unit being a 16 division of LP_DYN_SLOW_CLK. (R/W)

Register 14.65. PMU_EXT_LDO_P1_OP2A_ANA_REG (0x01DC)

PMU_ANA_OP2A_DREF_1			PMU_ANA_OP2A_EN_CUR_LIM_1			PMU_ANA_OP2A_EN_VDET_1			PMU_ANA_OP2A_MUL_1			(reserved)																							0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
31	28	27	26	25	23	22																															0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
0xb		0	0	0x2		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PMU_ANA_OP2A_MUL_1 Configures the MUL, where the output voltage of the regulator is $MUL \times V_{ref}$.

0-7: MUL ranges from 1 to 2.75, with a step of 0.25.

(R/W)

PMU_ANA_OP2A_EN_VDET_1 Configures whether to enable VO4 voltage detection.

0: Do not enable

1: Enable

(R/W)

PMU_ANA_OP2A_EN_CUR_LIM_1 Configures whether to enable over current protection for VO4.

0: Do not enable

1: Enable

(R/W)

PMU_ANA_OP2A_DREF_1 Configures the LDO V_{ref} , where the output voltage of the regulator is $MUL \times V_{ref}$.

0-8: $V_{ref} = 0.5\text{ V} - 0.9\text{ V}$, with a step of 50 mV

9-15: $V_{ref} = 1.0\text{ V} - 1.6\text{ V}$, with a step of 100 mV

(R/W)

Register 14.66. PMU_EXT_WAKEUP_LV_REG (0x01E8)

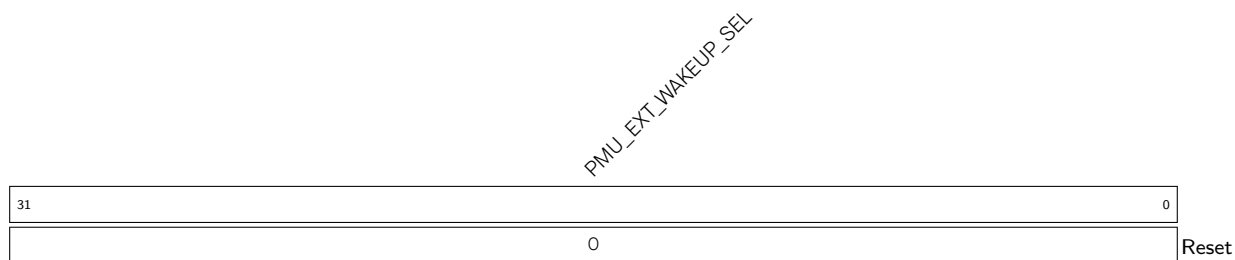
PMU_EXT_WAKEUP_LV																																0		
31																																		0
0																																		Reset

PMU_EXT_WAKEUP_LV Configures the wake level for EXT wake-up.

0: Low level wake-up

1: High level wake-up

(R/W)

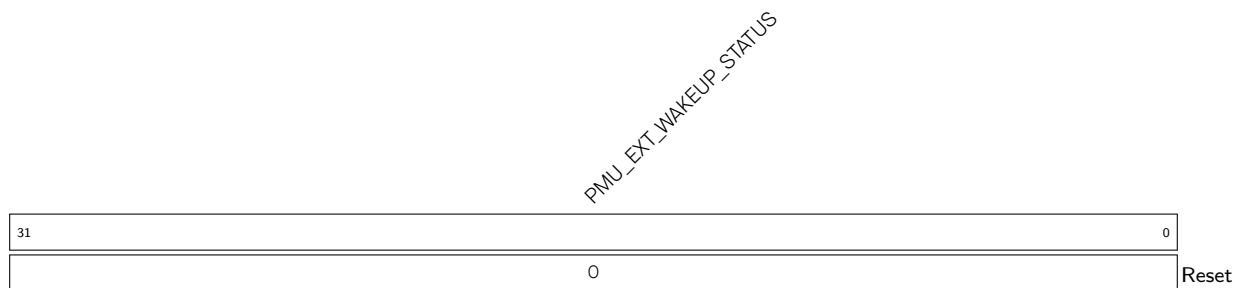
Register 14.67. PMU_EXT_WAKEUP_SEL_REG (0x01EC)

PMU_EXT_WAKEUP_SEL Configures whether to enable EXT wake-up.

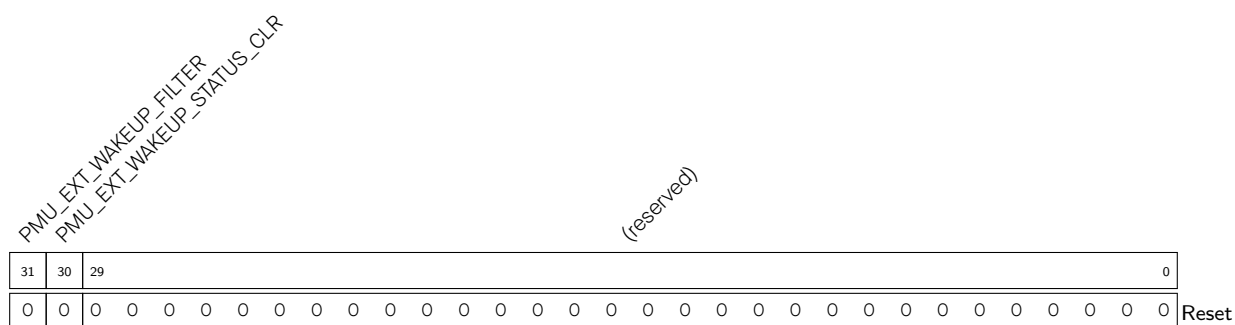
0: Disable

1: Enable

(R/W)

Register 14.68. PMU_EXT_WAKEUP_ST_REG (0x01F0)

PMU_EXT_WAKEUP_STATUS Gets the GPIO number for wake-up. 0 corresponds to GPIO0, 1 corresponds to GPIO1, and so on. (RO)

Register 14.69. PMU_EXT_WAKEUP_CNTL_REG (0x01F4)

PMU_EXT_WAKEUP_STATUS_CLR Clears the wake-up records. (R/W)

PMU_EXT_WAKEUP_FILTER Configures whether to enable the wake-up filter.

0: Disable

1: Enable

(R/W)

Register 14.70. PMU_SDIO_WAKEUP_CNTL_REG (0x01F8)

(reserved)										PMU_SDIO_ACT_DNUM									
31										9									
0 0										0x3ff									
										Reset									

PMU_SDIO_ACT_DNUM Configures the duration for maintaining the working state of SDIO. The unit is LP_DYN_FAST_CLK. (R/W)

Register 14.71. PMU_TOUCH_PWR_CNTL_REG (0x0210)

31		30		29						14		13				5		4				0		
0		0				100						10				0		0		0		0		Reset
PMU_TOUCH_SLEEP_TIMER_EN PMU_TOUCH_FORCE_DONE PMU_TOUCH_SLEEP_CYCLES PMU_TOUCH_WAIT_CYCLES (reserved)																								

PMU_TOUCH_WAIT_CYCLES Configures the waiting time for the PMU to wake up from the TOUCH timer, with the unit of LP_DYN_SLOW_CLK. (R/W)

PMU_TOUCH_SLEEP_CYCLES Configure the sleep time for TOUCH, with the unit of LP DYN SLOW CLK. (R/W)

PMU_TOUCH_FORCE_DONE Force return to the working state of the TOUCH timer. (R/W)

PMU_TOUCH_SLEEP_TIMER_EN Enables the TOUCH timer to start working. (R/W)

Register 14.72. PMU_DATE_REG (0x03FC)

31	30	0
0	0x2303140	

PMU_CLK_EN

PMU_PMU_DATE

Reset

PMU_PMU_DATE Version control register. (R/W)

PMU_CLK_EN Force enables the automatic clock gating of the register. (R/W)